ETHERNET-

MACs, Interfaces, & Mgmt Controllers Overview

w/ Debug/Bring-Up Tips

For QorIQ Product Family (P, T, LS, LA, & LX Series)

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BL Edge Processing
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SECURE CONNECTIONS
FOR A SMARTER WORLD

- The content in this presentation is focused on the Ethernet related interfaces supported by the Digital Networking (DN) Group at NXP Semiconductors.

- The presentation is NOT a deep dive of the architecture but merely an overview of the implementations and Ethernet startup hurdles that may be encountered when designing with the DN product family.

- Overview
- Parallel Data Interfaces
- Serial Data Interfaces
- PHYless Considerations
- Media Access Controllers (MACs)
- MII Management Bus and Controller (Two Wires: MDC & MDIO)
- Signal Integrity Considerations
- Documentation : SoC and DPAA2 Reference Manuals
- Development Boards
- U-Boot command line check for link status
- Summary / Conclusion

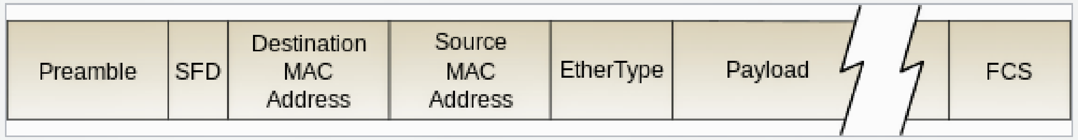
Overview

SECURE CONNECTIONS
FOR A SMARTER WORLD

Ethernet packet – physical layer [edit]

Preamble and start frame delimiter [edit]

See also: Syncword



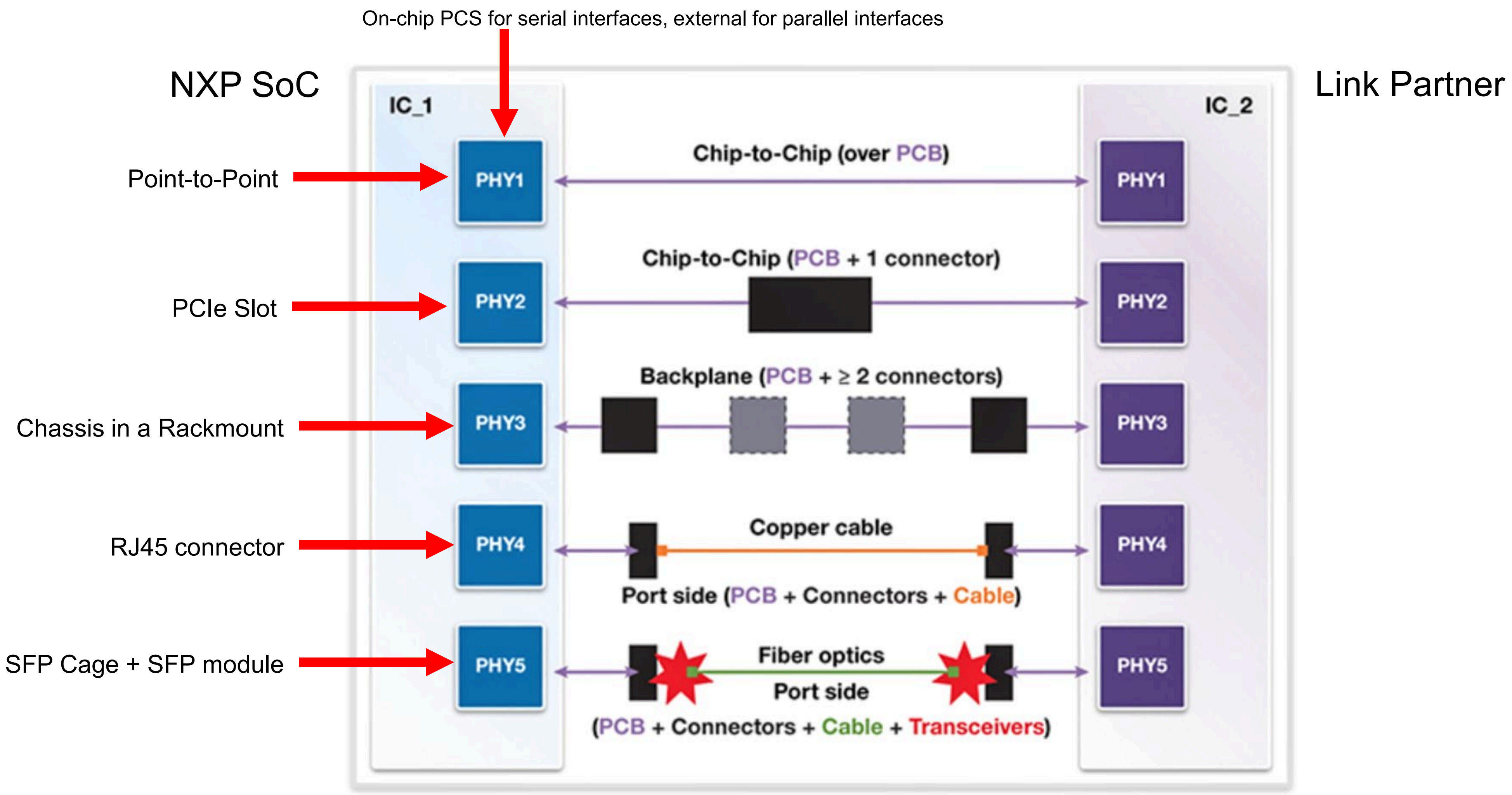
An Ethernet frame inside an Ethernet packet, with SFD marking the end of the packet preamble and indicating the beginning of the frame.[3]

An Ethernet packet starts with a seven-octet preamble and one-octet *start frame delimiter* (SFD).[c]

The preamble consists of a 56-bit (seven-byte) pattern of alternating 1 and 0 bits, allowing devices on the network to easily synchronize their receiver clocks, providing bit-level synchronization. It is followed by the SFD to provide byte-level synchronization and to mark a new incoming frame. For Ethernet variants transmitting serial bits instead of larger symbols, the (uncoded) on-the-wire bit pattern for the preamble together with the SFD portion of the frame is 10101010 10101010 10101010 10101010 10101010 10101010 10101010 10101011;[3]:sections 4.2.5 and 3.2.2 The bits are transmitted in order, from left to right.[3]:sections 4.2.5

The SFD is the eight-bit (one-byte) value that marks the end of the preamble, which is the first field of an Ethernet packet, and indicates the beginning of the Ethernet frame. The SFD is designed to break the bit pattern of the preamble and signal the start of the actual frame.[3]:section 4.2.5 The SFD is immediately followed by the destination MAC address, which is the first field in an Ethernet frame. SFD is the binary sequence 10101011 (0xAB, decimal 171 in the Ethernet LSB first bit ordering).[3]:sections 3.2.2, 3.3 and 4.2.6

Physical layer transceiver circuitry (PHY for short) is required to connect the Ethernet MAC to the physical medium. The connection between a PHY and MAC is independent of the physical medium and uses a bus from the media independent interface family (MII, GMII, RGMII, SGMII, XGMII). Fast Ethernet transceiver chips utilize the MII bus, which is a four-bit (one nibble) wide bus, therefore the preamble is represented as 14 instances of 0xA, and the SFD is 0xA 0xB (as nibbles). Gigabit Ethernet transceiver chips use the GMII bus, which is an eight-bit wide interface, so the preamble sequence followed by the SFD would be 0xAA 0xAA 0xAA 0xAA 0xAA 0xAA 0xAA 0xAB (as bytes).



SOURCE: https://www.synopsys.com/designware-ip/technical-bulletin/ethernet-dwtb-q117.html

SPEED	MEDIUM	CONNECTOR	MAC INTERFACE
1 Gbps	1000BASE-T	RJ45	RGMII
1 Gbps	1000BASE-X	SFP	1000BASE-X SGMII USXGMII
1 Gbps	Backplane (1m)	pcb trace, interconnect	1000BASE-KX
2.5 Gbps	2500BASE-X	SFP	OC-SGMII USXGMII
10 Gbps	10GBASE-T	RJ45	XFI / XAUI USXGMII
10 Gbps	10GBASE-R	SFP+ / QSFP	XFI[1] / SFI / XAUI USXGMII
10 Gbps	Backplane (1m)	pcb trace, interconnect	10GBASE-KR
40 Gbps	40GBASE-R/C2C/C2M	QSFP+	XLAUI
40 Gbps	Backplane (1m)	pcb trace, interconnect	40GBASE-KR4
25 Gbps	25GBASE-R	SFP28	25GAUI
25 Gbps	Backplane (1m)	pcb trace, interconnect	*25GBASE-KR
50 Gbps	50GBASE-R	SFP28	50GAUI-2
100 Gbps	100GBASE-R/C2C/C2M	QSFP28	CAUI-4

TIP: PHYless configurations also feasible if simulation/modeling yield positive results.

[1]XFI may require a retimer/repeater to convert XFI to SFI if the SFP module only supports SFI electricals

The NXP QorIQ product families support the Ethernet protocol with a variety of controllers for the data path and the PCS/PHY management path.

Parallel data interfaces are MII, RMII, GMII, and RGMII.

Serial data interfaces are SGMII, OC-SGMII (Overclocked), QSGMII, XAUI, XFI,SFI, USXGMII, XLAUI, 25GAUI, 50GAUI-2, CAUI-4 (with some backplane implementations as well).

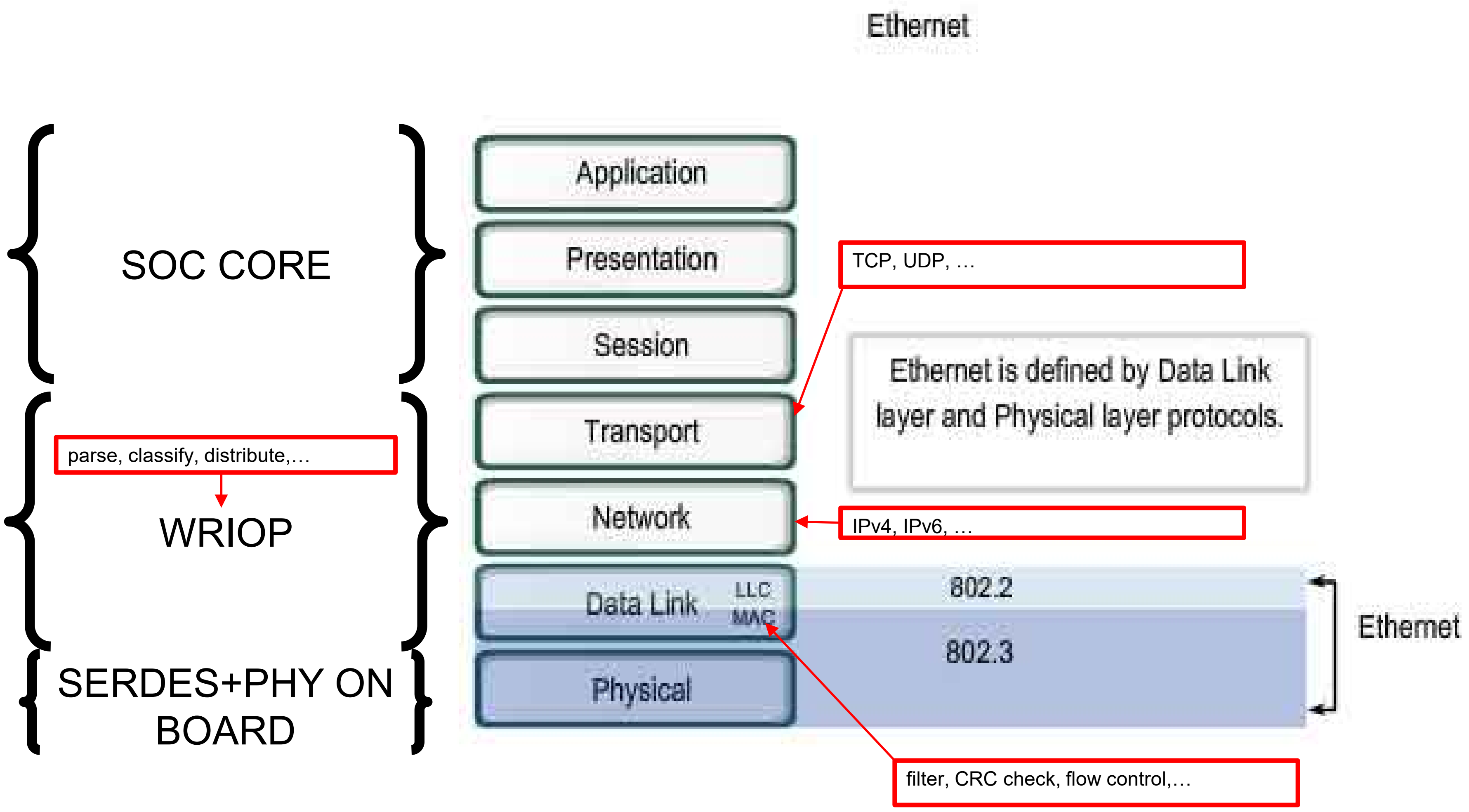
For the P-series, the Ethernet controllers are DPAA1-FMAN-dTSEC and DPA11-FMAN-10GEC.

For the T-series, the main Ethernet controller is DPAA1-FMAN-mEMAC.

For the LS-series, the main Ethernet controllers are eTSEC 2.x, PPFE, DPAA1-FMAN-mEMAC, ENETC, and DPAA2-WRIOP-mEMAC.

For the LA-series, the main Ethernet controllers are ??NETC?? and DPAA2-WRIOP-mEMAC.

For the LX-series, the main Ethernet controllers are DPAA2-WRIOP-mEMAC and DPAA2-WRIOP-cEMAC (for ≥ 40GbE).


Ethernet
SOC CORE
parse, classify, distribute,…
WRIOP
SERDES+PHY ON BOARD
Application
Presentation
Session
Transport
Network
Data Link
LLC
MAC
Physical
TCP, UDP, …
Ethernet is defined by Data Link layer and Physical layer protocols.
IPv4, IPv6, …
802.2
802.3
Ethernet
filter, CRC check, flow control,…

OVERVIEW - SUPPORTED LAYERS OF THE PROTOCOL STACK BY NXP ETHERNET HW

- OSI Model Layers

Host Layers	7: Application end use (FTP, SMTP,…)
	6: Presentation encryption & decryption
	5: Session interhost communication
	4: Transport reliability & flow control
Media Layers	3: Network / Internet logical addressing
	2: Data Link (MAC) physical addressing
	1: Physical (PHY) transmission/reception of bits

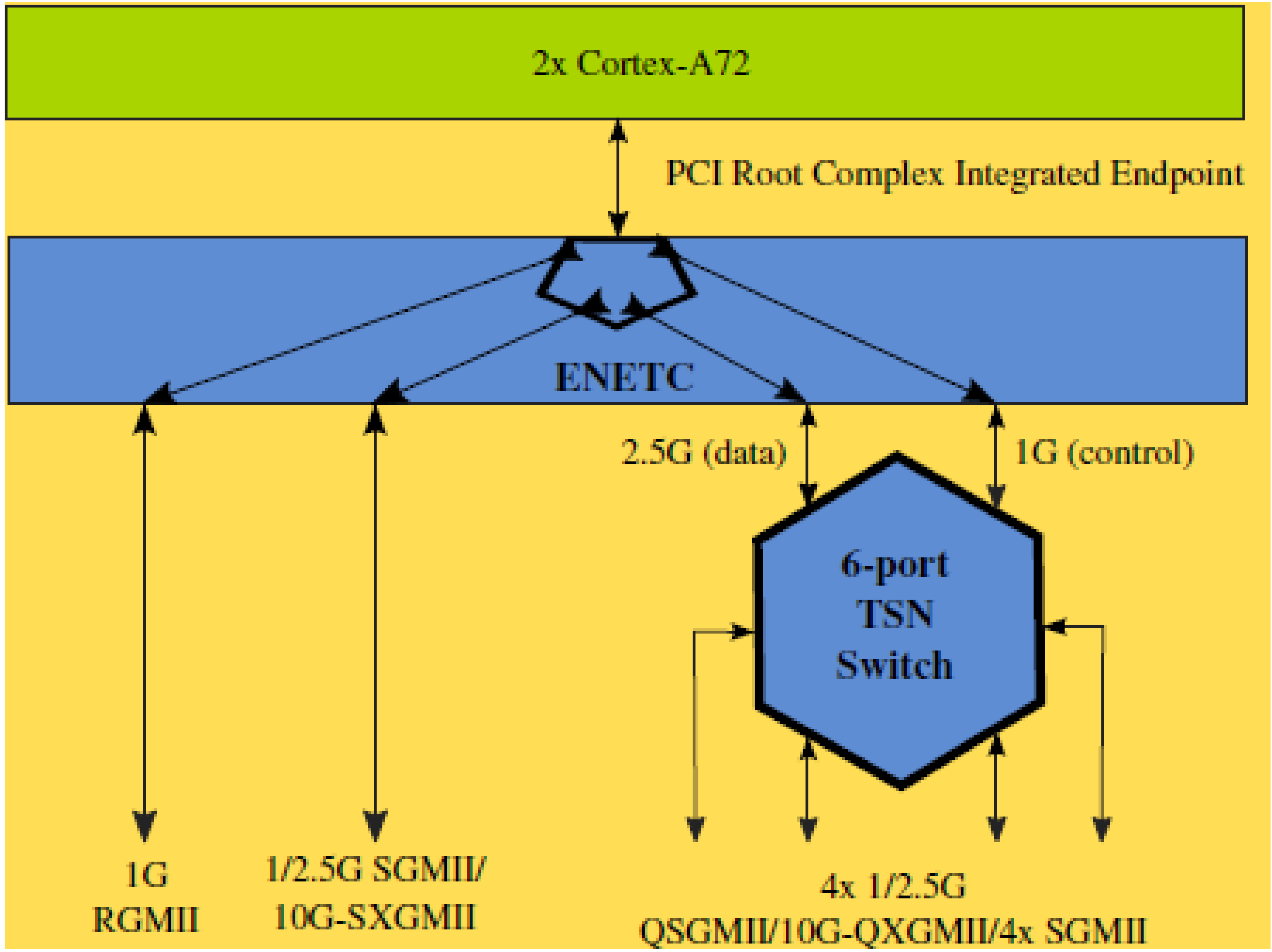
- Media Access Control (MAC)
  - Ethernet Type II Frame

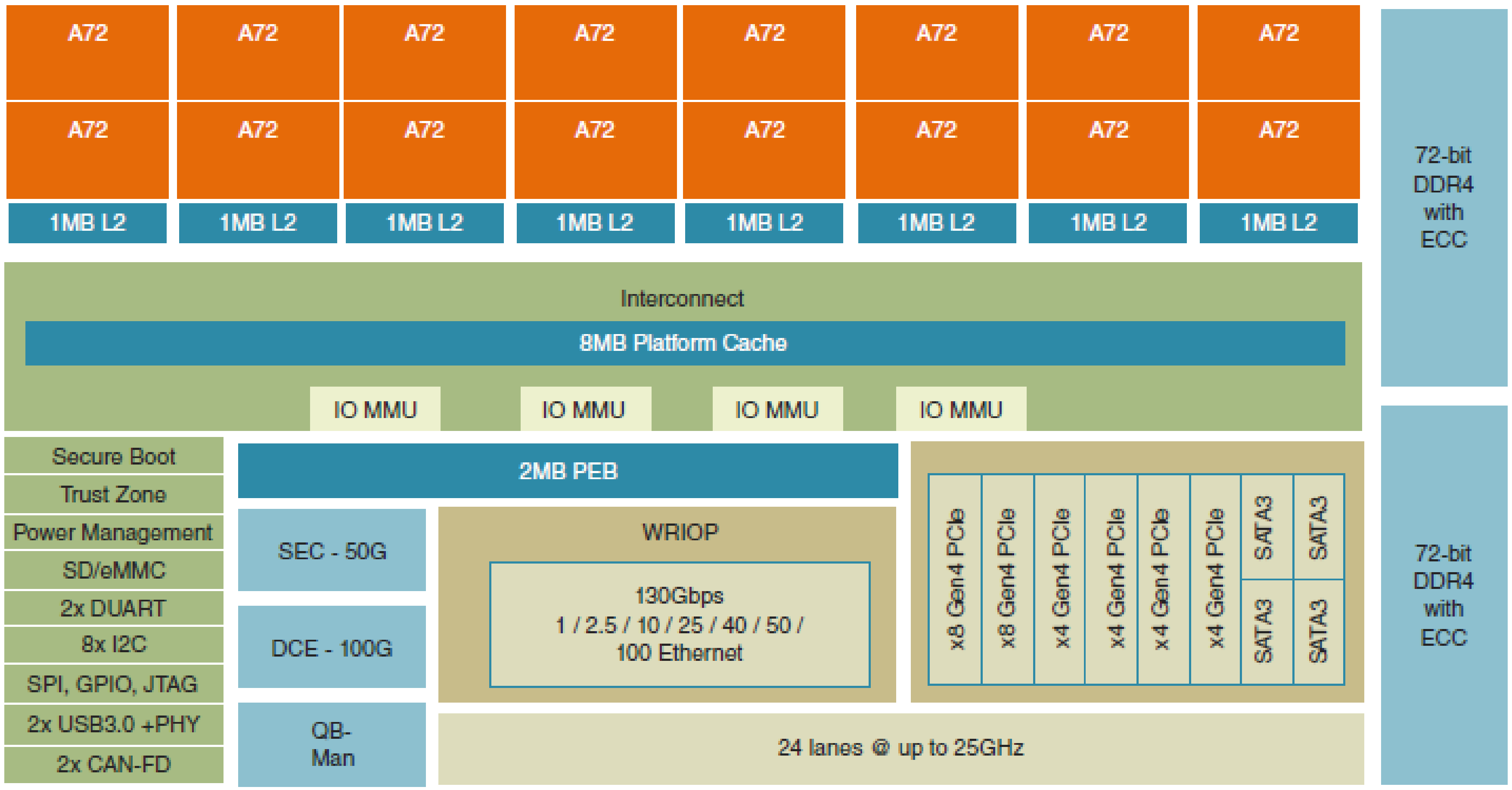
L3 header (IPv4/IPv6)

L4 header (TCP/UDP)

L2 header

Preamble	SFD	DA	SA	EthType	Payload	CRC32	IFG
7 bytes	1 byte	6 bytes	6 bytes	2 bytes	46 – 1500 bytes	4 bytes	12 bytes


2x Cortex-A72
PCI Root Complex Integrated Endpoint
ENETC
2.5G (data)
1G (control)
6-port
TSN
Switch
1G
RGMII
1/2.5G SGMII/
10G-SXGMII
4x 1/2.5G
QSGMII/10G-QXGMII/4x SGMII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1MB L2
1MB L2
1MB L2
1MB L2
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1MB L2
1MB L2
1MB L2
72-bit DDR4 with ECC
Interconnect
8MB Platform Cache
IO MMU
IO MMU
IO MMU
IO MMU
Secure Boot
Trust Zone
Power Management
SD/eMMC
2x DUART
8x I2C
SPI, GPIO, JTAG
2x USB3.0 +PHY
2x CAN-FD
2MB PEB
SEC - 50G
DCE - 100G
QB-Man
WRIOP
130Gbps
1 / 2.5 / 10 / 25 / 40 / 50 / 100 Ethernet
x8 Gen4 PCIe
x8 Gen4 PCIe
x4 Gen4 PCIe
x4 Gen4 PCIe
x4 Gen4 PCIe
x4 Gen4 PCIe
SATA3
SATA3
SATA3
SATA3
24 lanes @ up to 25GHz
72-bit DDR4 with ECC

Table 11-1. MAC Capabilities

MAC	Base Address	RGMII 1 Gbps	SGMII 1 Gbps	XFI 10 Gbps	10G-SXGMII	25 Gbps	50/40 Gbps	MACSEC Supported
1	8C0_7000h	—	—	—	—	—	Y	—
2	8C0_B000h	—	—	—	—	—	Y	—
3	8C0_F000h	—	Y	Y	Y	Y	—	Y
4	8C1_3000h	—	Y	Y	Y	Y	—	Y
5	8C1_7000h	—	Y	Y	Y	Y	—	Y
6	8C1_B000h	—	Y	Y	Y	Y	—	Y
7	8C1_F000h	—	Y	Y	Y	—	—	—
8	8C2_3000h	—	Y	Y	Y	—	—	—
9	8C2_7000h	—	Y	Y	Y	Y	—	—
10	8C2_B000h	—	Y	Y	Y	Y	—	—
11	8C2_F000h	—	Y	—	—	—	—	—
12	8C3_3000h	—	Y	—	—	—	—	—
13	8C3_7000h	—	Y	Y	Y	—	—	—
14	8C3_B000h	—	Y	Y	Y	—	—	—
15	8C3_F000h	—	Y	—	—	—	—	—
16	8C4_3000h	—	Y	—	—	—	—	—
17	8C4_7000h	Y (EC1)	Y	—	—	—	—	—
18	8C4_B000h	Y (EC2)	Y	—	—	—	—	—

The formula for the specific DPAA2 MAC controller register offset in CCSR space is

8C0_0000 + (hexadecimal Mac Port #)*(0x4000) + 0x3000.

Hence

MAC1 registers @ 8C0_7000
MAC2 registers @ 8C0_B000
MAC3 registers @ 8C0_F000
MAC4 registers @ 8C1_3000
MAC5 registers @ 8C1_7000
MAC6 registers @ 8C1_B000
MAC7 registers @ 8C1_F000
MAC8 registers @ 8C2_3000
MAC9 registers @ 8C2_7000
MAC10 registers @ 8C2_B000
MAC11 registers @ 8C2_F000
MAC12 registers @ 8C3_3000
MAC13 registers @ 8C3_7000
MAC14 registers @ 8C3_B000
MAC15 registers @ 8C3_F000
MAC16 registers @ 8C4_3000
MAC17 registers @ 8C4_7000
MAC18 registers @ 8C4_B000

For external MDIO, we use offsets for EMI1 and EMI2, but programming is the same.
EMI1 registers @ 8B9_6000
EMI2 registers @ 8B9_7000

Multirate Ethernet Media Access Controller (mEMAC)

Table 12-4. Module Memory Map (continued)

mEMAC Offset[1]	Name	Access	Reset	Section/Page
0x00C	MAC_ADDR_0—Lower 32 bits of the first 48-bit MAC address	R/W	0x0000_0000	12.4.3.1.2/12-12
0x010	MAC_ADDR_1—Upper 16 bits of the first 48-bit MAC address	R/W	0x0000_0000	12.4.3.1.3/12-12
0x014	MAXFRM—Maximum frame length register	R/W	0x0000_0600	12.4.3.1.4/12-13
0x01C	RX_FIFO_SECTIONS—Receive FIFO configuration register	R/W	0x0000_0020	12.4.3.1.5/12-13
0x020	TX_FIFO_SECTIONS—Transmit FIFO configuration register	R/W	Device-specific	12.4.3.1.6/12-14
0x040	IEVENT—Interrupt event register	R/W	0x0000_8060	12.4.3.1.7/12-16
0x044	TX_IPG_LENGTH—Transmitter inter-packet-gap register	R/W	0x0000_000C	12.4.3.1.8/12-17
0x054	CL01_PAUSE_QUANTA - CL0,1 Pause quanta register	R/W	0x0000_0000	12.4.3.1.10/12-19
0x058	CL23_PAUSE_QUANTA - CL2,3 Pause quanta register	R/W	0x0000_0000	12.4.3.1.11/12-19
0x05C	CL45_PAUSE_QUANTA - CL4,5 Pause quanta register	R/W	0x0000_0000	12.4.3.1.12/12-19
0x060	CL67_PAUSE_QUANTA - CL6,7 Pause quanta register	R/W	0x0000_0000	12.4.3.1.13/12-20
0x064	CL01_PAUSE_THRESH - CL0,1 Pause quanta threshold register	R/W	0x0000_0000	12.4.3.1.14/12-20

TIP: Each MAC has the same programming model. Use offsets to access the registers. Some configuration registers should not be changed on-the-fly.

12.4.3.1 mEMAC General Control and Status Registers

This section describes general control and status registers used for both transmitting and receiving Ethernet frames. All of the registers are 32 bits wide.

12.4.3.1.1 Command and Configuration Register (COMMAND_CONFIG)

Offset 0x008 (COMMAND_CONFIG)
Offset_from (PhysicalPort)

Access: Mixed

Bit	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
R W	MG		RXSTP	-	FLT_H-DL_DIS	—		REG_LOWP_RXETY	TX_LOWP_ENA	—	SFD	—	PFC_MODE	—		SEND_IDLE
Reset	In table															

Bit	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
R		—	CNT_FRM_EN	SWR	TXP	XGLP	—	PAUSE_IGN	PAUSE_FWD	FCS	—	PROMIS	—	FUTURE5	RX_EN	TX_EN
W				w1c												
Reset	In table															

Figure 12-2. Command and Configuration Register (COMMAND_CONFIG)

TIP: Setting promiscuous mode to accept all packets is sometimes useful for debug.

Multirate Ethernet Media Access Controller (mEMAC)

12.4.3.1.7 Interrupt Event Register (IEVENT)

The IEVENT reflects the occurrence of various errors or functional events. All bits are cleared by writing 1 except for bits 16, 25, 26, and 27 which are status bits.

Offset 0x040 (IEVENT) Access: Mixed
Offset_from (PhysicalPort)

Bits	0–15
R	PCS_EVENTS
W	w1c
Reset	All zeros

Bits	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
R	TX_IDLE	MGI	—	RX_OVFL	TX_UNFL	TX_OVFL	TX_ECC_E	RX_ECC_E	LI_FAULT	RX_EMPTY	TX_EMPTY	RX_LOWP	TS_AVAIL	—	REM_FAULT	LOC_FAULT
W		w1c		w1c	w1c	w1c	w1c	w1c	w1c				w1c	w1c	w1c	w1c
Reset	1	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0

Figure 12-8. Interrupt Event Register (IEVENT)

TIP: Cleared faults equate to transmitter OK to TX packets.

12.4.3.2.2 Receive Octets Counter Register (ROCT*n*)

Offset 0x108 (ROCTn)
Offset_from (PhysicalPort)

Access: Read only

	0 – 63
R	ROCT*n*
W	
Reset	All zeros

Figure 12-26. Receive Octets Counter Register (ROCT*n*)

Bits	Name	Description
0–63	ROCT*n*	Incremented for each octet received except preamble (that is, Header, Payload, Pad and FCS) for all valid frames and valid Pause frames received.

Table 12-32. ROCT*n* Field Description

TIP: Non-zero counters are a good indication packets are being received and sent.

12.4.3.2.27 Transmit Octets Counter Register (TOCT*n*)

Offset 0x208 (TOCTn)
Offset_from (PhysicalPort)

Access: Read only

	0 – 63
R	TOCT*n*
W	
Reset	All zeros

Figure 12-51. Transmit Octets Counter Register (TOCT*n*)

Bits	Name	Description
0–63	TOCT*n*	Incremented for each octet transmitted except preamble (that is, Header, Payload, Pad and FCS) for all valid frames and valid Pause frames transmitted.

Table 12-57. TOCT*n* Field Description

26.4.1 SerDes memory map

SerDes1 base address: 1EA_0000h

SerDes2 base address: 1EB_0000h

Offset	Register	Width (In bits)	Access	Reset value
0h	SerDes Reset Control Register (RSTCTL)	32	W1C	0000_0000h
2Ch	SerDes Left End Cap Control Register 3 (LCAPCR3)	32	RW	Table 26-3
400h	SerDes PLLn Reset Control Register (PLLFRSTCTL)	32	RW	Table 26-3
404h	SerDes PLLn Control/Status Register 0 (PLLFCR0)	32	RW	Table 26-3
408h	SerDes PLLn Control/Status Register 1 (PLLFCR1)	32	RW	Table 26-3
410h	SerDes PLLn Control/Status Register 3 (PLLFCR3)	32	RW	0000_0000h
414h	SerDes PLLn Control/Status Register 4 (PLLFCR4)	32	RW	0000_0000h

TIP: SerDes has block and per lane specific controls. Use offsets to access the registers.

26.4.1.18.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	RST_REQ	RST_DONE	Reserved		HLT_REQ	STP_REQ	Reserved	DIS	Reserved							
W	W1C				W1C	W1C										
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved			CDR_LOCK	Reserved											
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TIP: The CDR (Clock Data Recovery) bit indicates the lane can recover a clock.
Does NOT guarantee the data is valid.

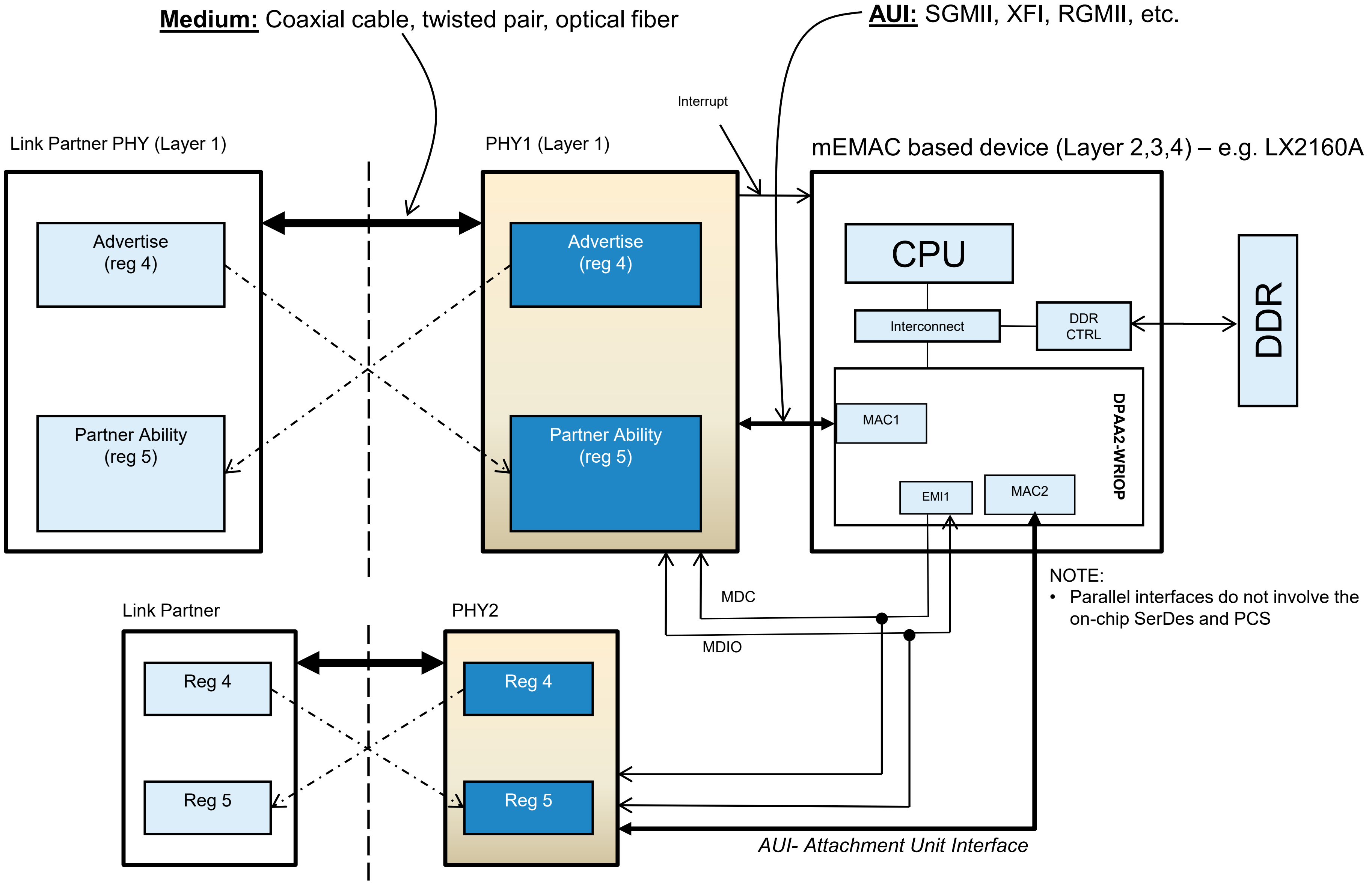
The Ethernet management controllers support either IEEE 802.3 Clause 22 or Clause 45 or both.

External PHY registers are accessed via a specified MAC or a dedicated controller (e.g. EMI1 or EMI2).

Internal PCS registers are accessed via registers in the associated MAC.

Clause 22 has been associated with 10/100/1000Mbps PHYs and Clause 45 with 10Gbps PHYs but this is not a hard restriction. It all depends on the implementation by the component manufacturer.

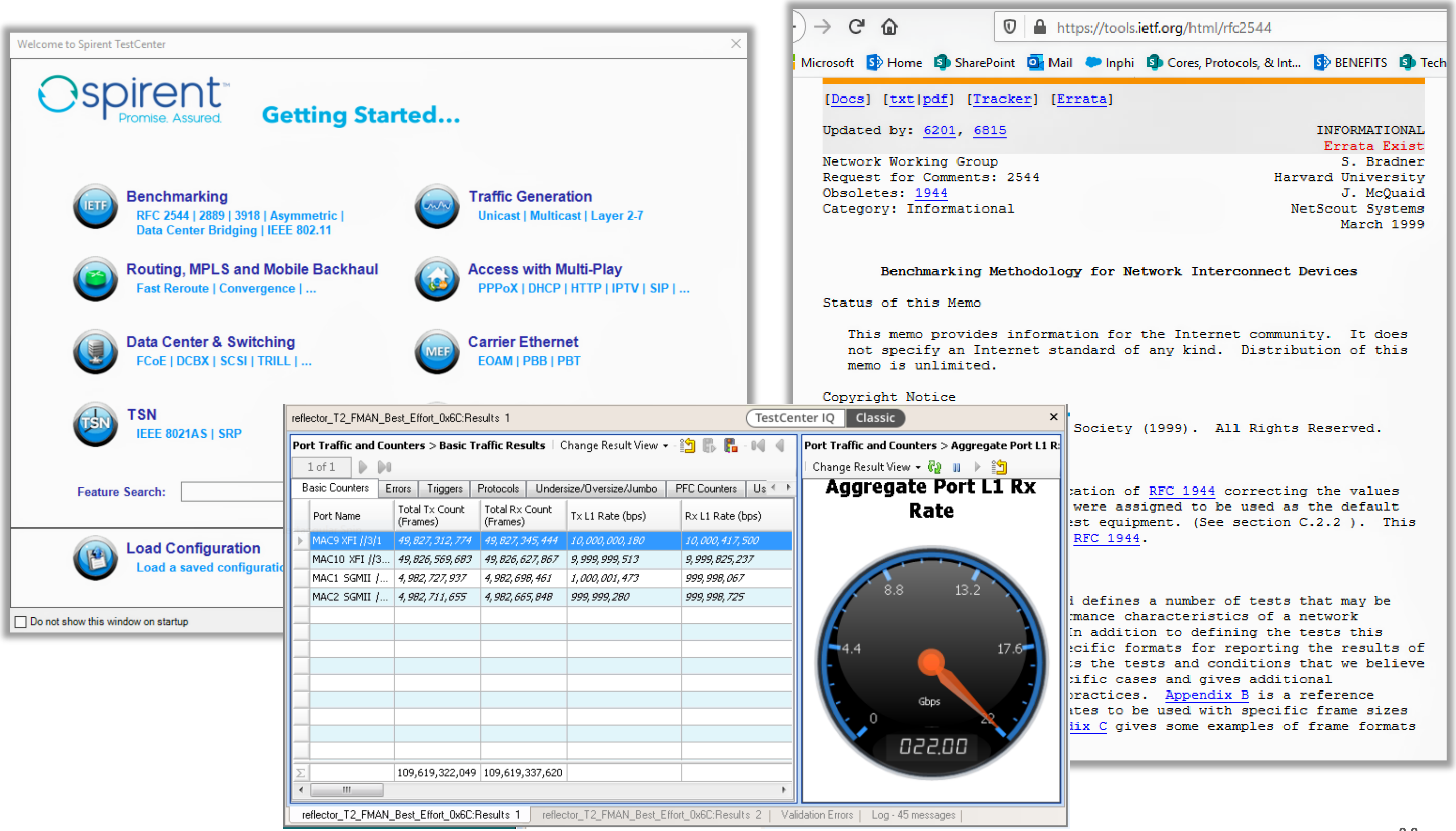
e.g., 1G PHYs with C45 Energy Efficient Ethernet (EEE) registers



NOTE:

- Parallel interfaces do not involve the on-chip SerDes and PCS

OVERVIEW – PACKET GENERATION AND ANALYSIS

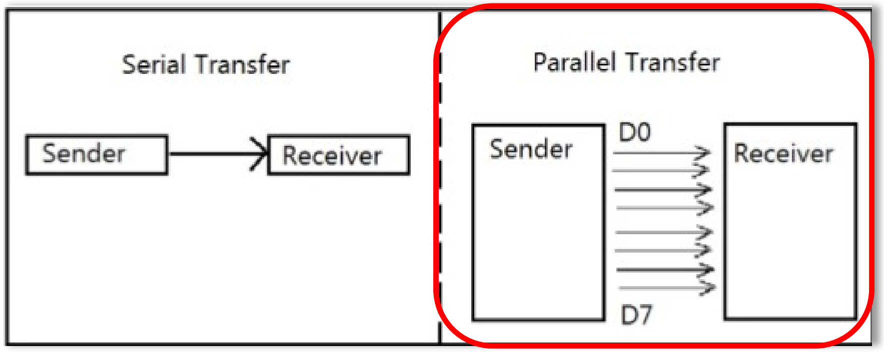


Parallel Data Interfaces

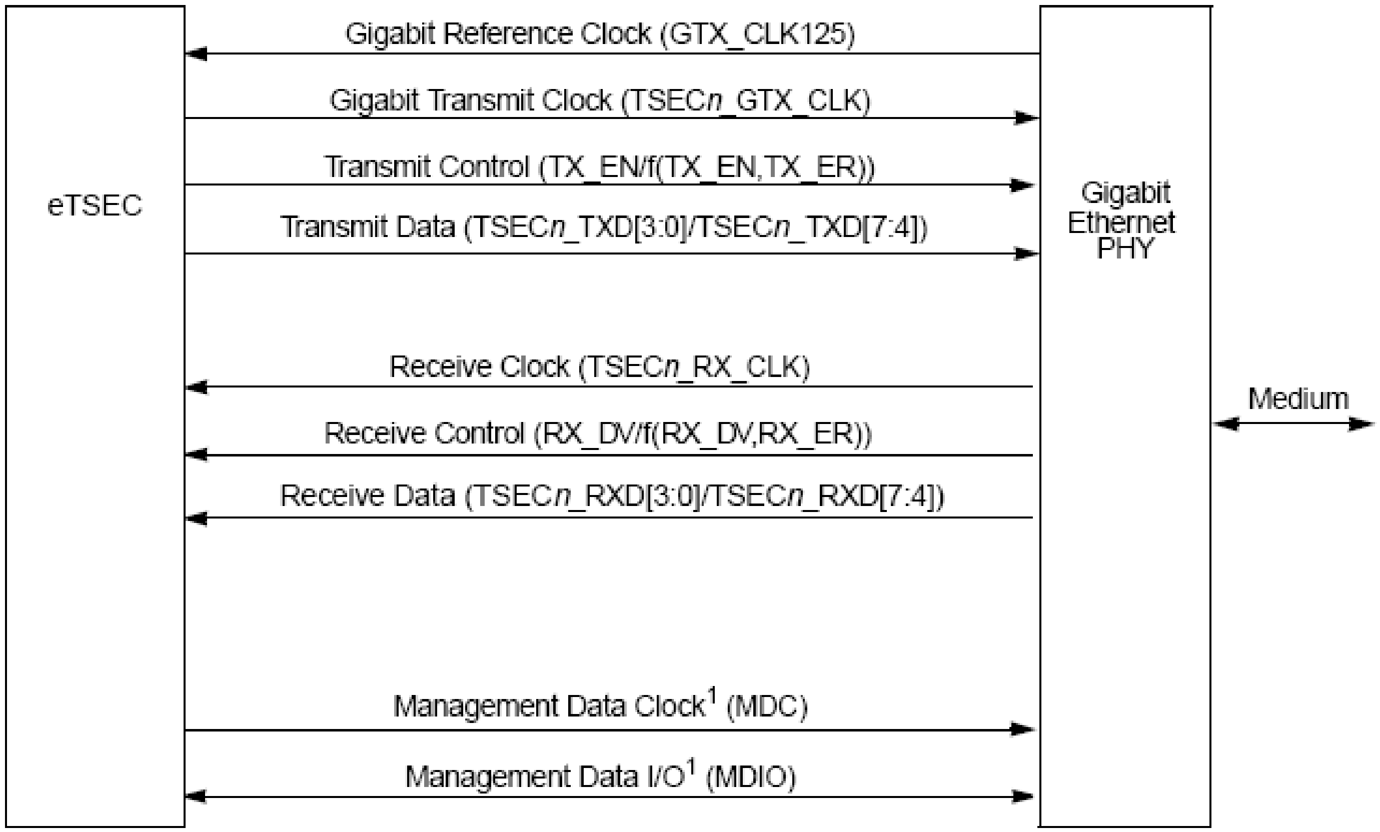
SECURE CONNECTIONS
FOR A SMARTER WORLD

- MII (10/100 Mbps)
  - Half-Duplex
  - Full-Duplex

- RGMII (10/100/1000 Mbps)
  - Half-Duplex
    - Supported Only DPAAx SoCs with the MII interface
  - Full-Duplex

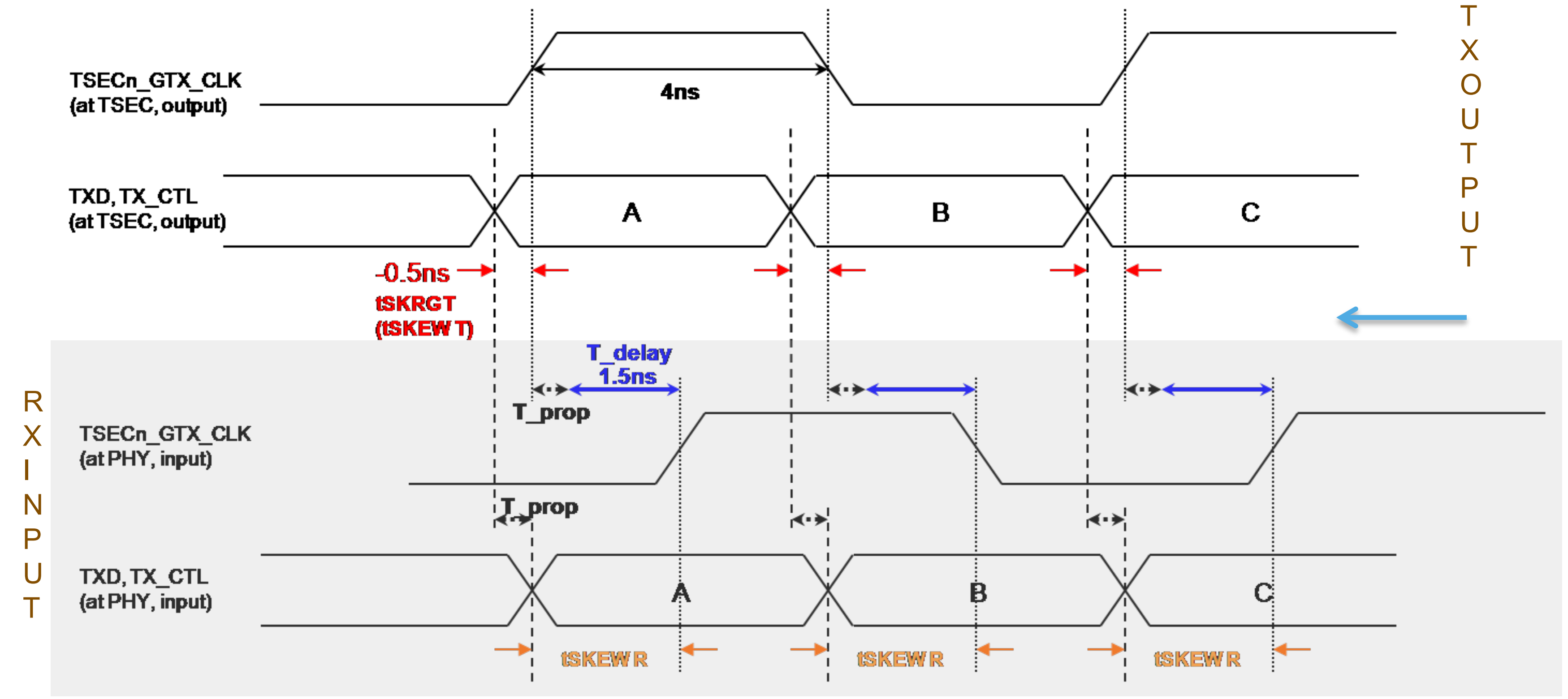


- External 125Mhz clock source is needed
- eTSEC outputs TX 125MHz clock
- eTSEC has internal dividers to support 100Mbps & 10Mbps
- Half-Duplex is encoded (No COL and CRS signals used)
- TX and RX have different timings for the data (needed delay provided by trace or PHY)

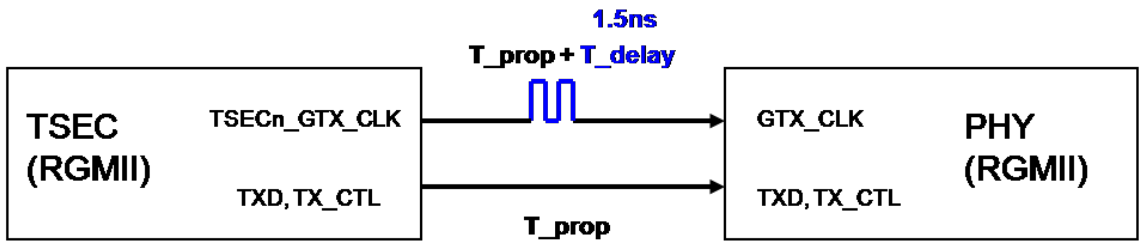


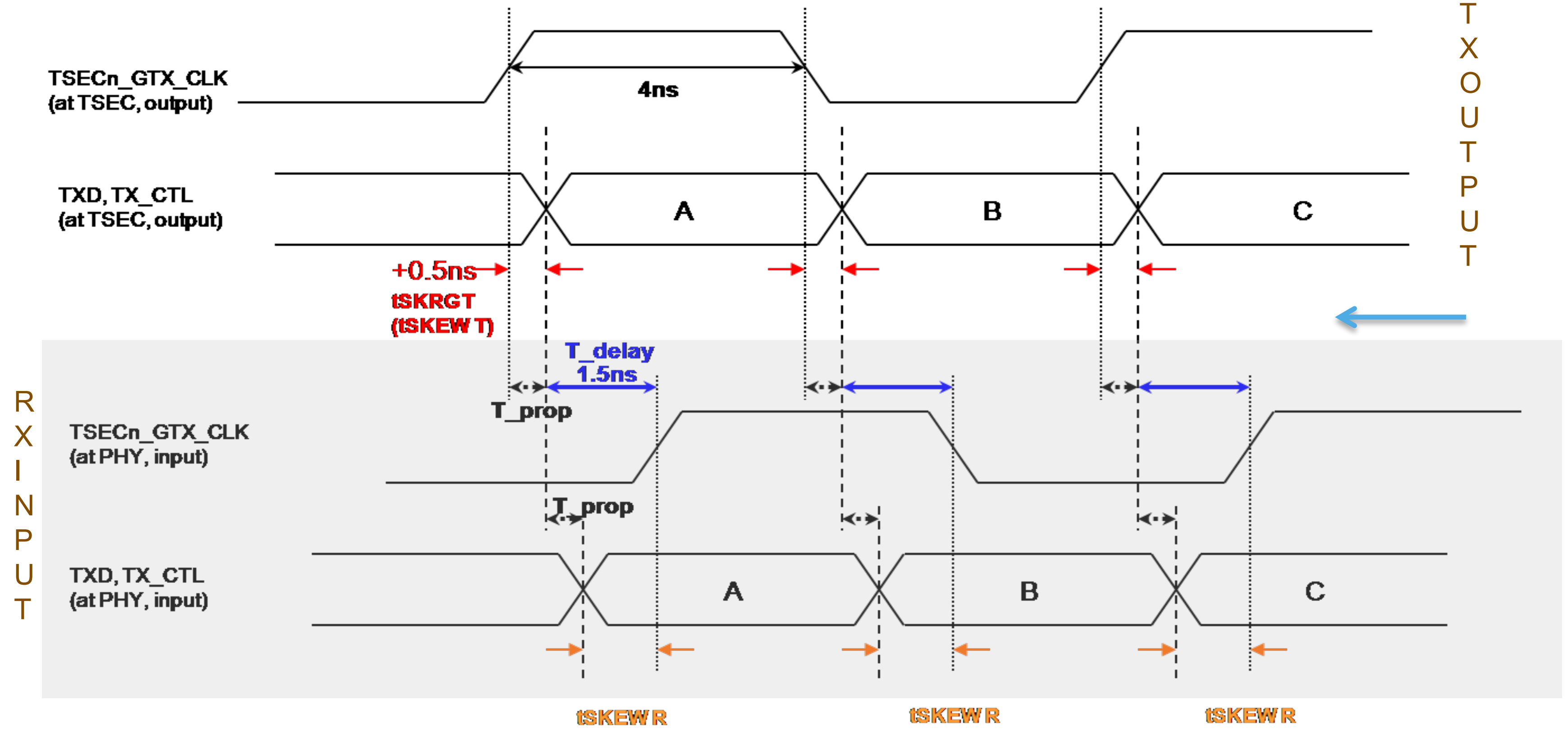
[1] The management signals (MDC and MDIO) are common to all of the gigabit Ethernet controllers module connections in the system, assuming that each PHY has a different management address.

PARALLEL DATA INTERFACES - TX DATA W.R.T. GTX_CLK (TSKEW T = -0.5NS)

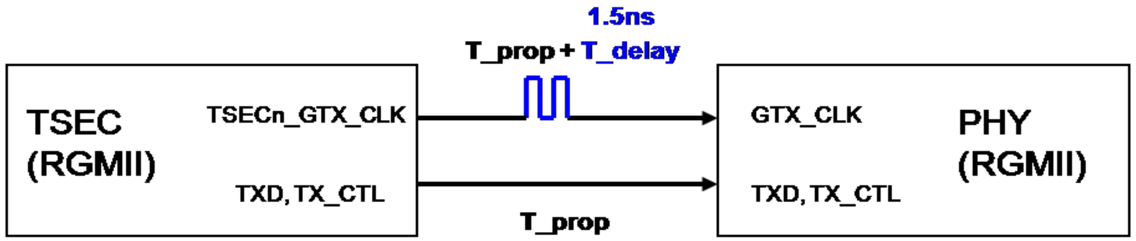


Spec	Min	max
tSKEW T	-0.5ns	0.5ns
tSKEW R	1ns	2.6ns





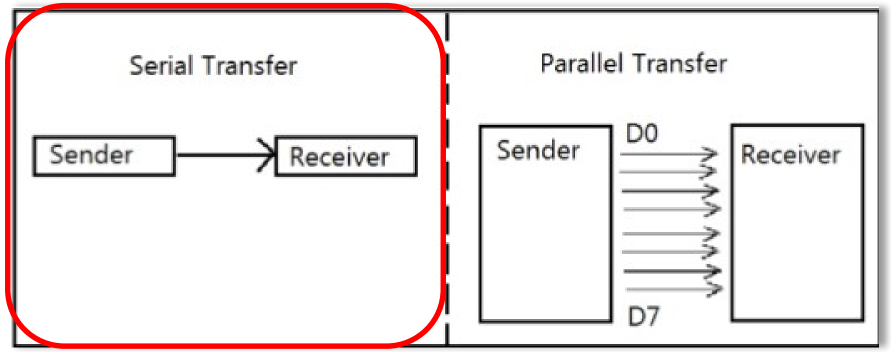
Spec	Min	max
tSKEW T	-0.5ns	0.5ns
tSKEW R	1ns	2.6ns



Serial Data Interfaces

SECURE CONNECTIONS
FOR A SMARTER WORLD

- SGMII (10/100/1000 Mbps)
  - 1.25Gbps, 8b/10b, SGMII ENG-46158 leveraging Clause 36
- XFI / SFI (10Gbps)
  - 10.3125Gbps SerDes lane, 64b/66b, Clause 49
- USXGMII (100M/1G/2.5G/~~5G~~/10G)
  - 10.3125Gbps SerDes lane, 64b/66b, Clause 49
- 25GAUI (25Gbps)
  - 25.78125Gbps SerDes lane, 64b/66b, Clause 107
- XLAUI (40Gbps)
  - x4 10.3125Gbps SerDes lanes, 64b/66b ea., Clause 82
- 50GAUI-2/CAUI-4 (50Gbps/100Gbps)
  - x2 or x4, 25.78125Gbps SerDes lanes, 64b/66b, Clause 82



29.1 The SerDes module as implemented on the chip

29.1.1 SerDes protocol notation

The following table shows the supported networking protocol options for SerDes module. The following notational conventions are used in the table:

- SGMII notation:
  - SGMII<m>-T
    - "m" indicates whether SGMII (1 lane @1.25 Gbaud or 1 lane @ 3.125 Gbaud)
    - T indicates all SGMII 2.5G-T and SGMII 1G-T protocols are connected to the 2.5G capable port0 of the Ethernet network controller (ENETC). Only one protocol can be used at any time because they all come from one port.
  - SGMII<m>-S
    - "m" indicates whether SGMII (1 lane @1.25 Gbaud or 1 lane @ 3.125 Gbaud)
    - S indicates all SGMII 2.5G-S and SGMII 1G-S protocols are connected to the TSN Switch MACs. They are not supported simultaneously with the multi-MAC protocols because they reuse the same TSN Switch MACs.
- QSGMII/10G-SXGMII/10G-QXGMII notation:
  - All MACs of the multi-MAC protocols (10G-QXGMII and QSGMII) are connected to the TSN Switch MACs.
  - 10G-SXGMII is connected to the 2.5G capable port0 of the ENETC.
- PCIe notation:
  - PCIe<m> x<width> indicates the PCIe controller that is used by the protocol selection and the associated width configuration. m indicates the PCIe controller number and x indicates the associated width configuration.

26.4 SerDes Protocol Notation

(view resource)

The following table shows the supported networking protocol options for SerDes modules 1, 2, and 3 The following notation conventions are used in the table:

- [Protocol].[MAC(s)] indicates which WRIOP MAC(s) are used by the protocol selection as shown in the examples below.
  - SGMII.3 indicates 1G SGMII (1 lane @ 1Gbps, 1.25 Gbaud) using WRIOP MAC3.
  - USXGMII / XFI.1 indicates USXGMII or XFI (1 lane @ 10Gbps,10.3125 Gbaud) using WRIOP MAC1.

  - 25GE.4 indicates 25G Ethernet (1 lane @ 25Gbps, 25.39 Gbaud) using WRIOP MAC4.
  - 40GE.2 indicates 40G Ethernet (4 lanes @ 10Gbps, 10.3125 Gbaud) using WRIOP MAC2.
  - 50GE.1 indicates 50G Ethernet (2 lanes @ 25Gbps, 25.78125 Gbaud) using WRIOP MAC1.
  - 100GE.1 indicates 100G Ethernet (4 lanes @ 25Gbps, 25.78125 Gbaud) using WRIOP MAC1.

SerDes 1

The following table shows the supported SerDes 1 protocol combinations where the leftmost column is the RCW field used to select the protocol (SRDS_PRTCL_S1)

RCW[SRDS_PRTCL_S1] (decimal)	H Lane 0	G Lane 1	F Lane 2	E Lane 3	PLL mapping	PLL mapping after gen3/4 speed switch
0	off	off	off	off	SSSS	N/A
1	PCIe.1 x4				SSSS	FFFF
2	SGMII.3	SGMII.4	SGMII.5	SGMII.6	SSSS	N/A
3	USXGMII / XFI.3	USXGMII / XFI.4	USXGMII / XFI.5	USXGMII / XFI.6	SSSS	N/A
9	PCIe.1 x1	SGMII.4	SGMII.5	SGMII.6	SSSS	FSSS
11	PCIe.1 x2		SGMII.5	SGMII.6	SSSS	FFSS
15	50GE.1		50GE.2		FFFF	N/A
16	50GE.1		25GE.5	25GE.6	FFFF	N/A
17	25GE.3	25GE.4	25GE.5	25GE.6	FFFF	N/A
18	USXGMII / XFI.3	USXGMII / XFI.4	25GE.5	25GE.6	SSFF	N/A
20	40GE.1				SSSS	N/A

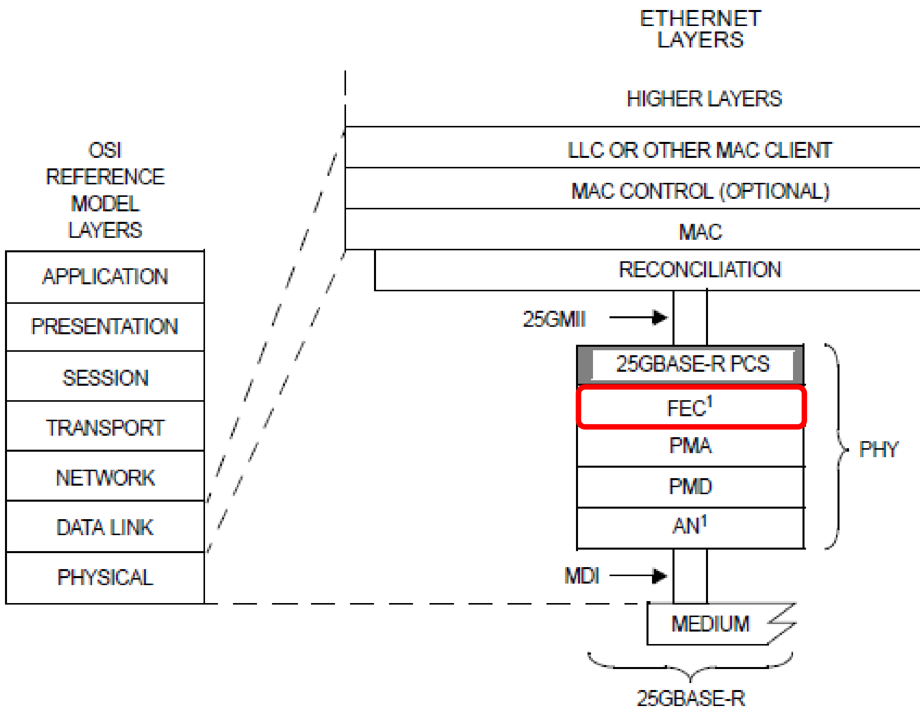
NOTES:
- 645MHz minimum platform frequency for 25GbE operation
- No spread-spectrum clocking support
- Check u-boot splash screen to confirm SerDes protocol selection

Table 26-2. Valid SerDes RCW Encodings and Reference Clocks

SerDes Protocol (given lane)	Valid Reference Clock Frequency	Valid setting as determined by SRDS_PRTCL_Sn	Valid Setting as Determined by SRDS_PLL_REF_CLK_SEL_Sn	Valid Setting as Determined by SRDS_DIV_*_Sn
Networking Interfaces				
SGMII (1.25 Gbps)	100 MHz	SGMII @ 1.25 Gbps	0 100 MHz	Don't care
	125 MHz		1 125 MHz	
USXGMII/XFI ,40GE (each lane: 10.3125 Gbaud)	156.25 MHz	USXGMII/XFI, 40GE @ 10.3125 Gbaud	0 156.25 MHz	Don't care
	161.1328 MHz		1 161.1328125 MHz	
25GE, 50GE, 100GE (each lane: 25.78125 Gbaud)	161.1328 125 MHz	25GE, 50GE, 100GE @ 25.78125 Gbaud	0 161.1328125 MHz	Don't care

1. A spread-spectrum reference clock is permitted for PCI Express. However, if any other high speed interface such as SGMII is used concurrently on the same SerDes bank, spread-spectrum clocking is not permitted.

IEEE Std 802.3by-2016
IEEE Standard for Ethernet—Amendment 2: Media Access Control Parameters, Physical Layers, and Management Parameters for 25 Gb/s Operation



25GMII = 25 GIGABIT MEDIA INDEPENDENT INTERFACE
AN = AUTO-NEGOTIATION
FEC = FORWARD ERROR CORRECTION
LLC = LOGICAL LINK CONTROL
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE
PCS = PHYSICAL CODING SUBLAYER

PHY = PHYSICAL LAYER DEVICE
PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT

NOTE 1—CONDITIONAL BASED ON PHY TYPE

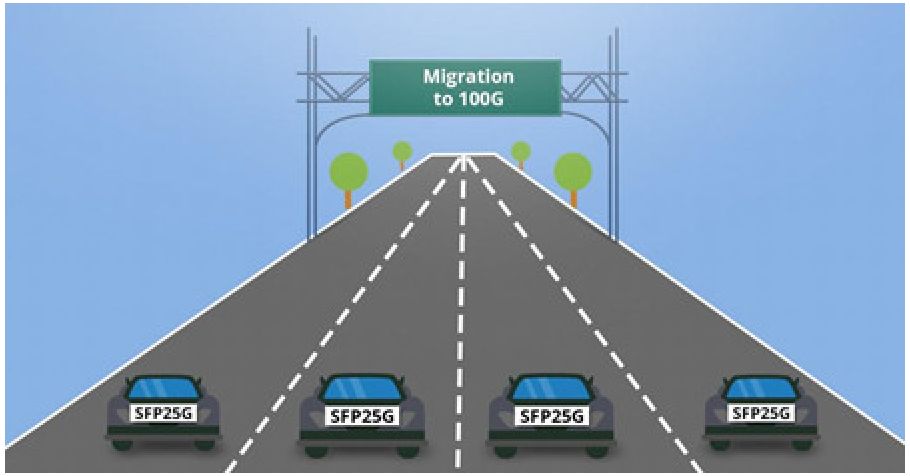
Figure 107–1—25GBASE-R PCS relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and IEEE 802.3 Ethernet model

NOTES:

- Forward Error Correction (FEC) gives the receiver the ability to correct errors without needing a reverse channel to request re-transmission of data.
- Main SoC components are Serializer-DeSerializer(SerDes), Physical Coding Sublayer (PCS), Protocol Converter (PCVTR), and Media Access Controller (MAC) to support the physical layer.

TIP: LX2 has options for FC-FEC (Clause 74) and **RS-FEC (Clause 91)** in IEEE 802.3.


optech
OPAX-S10-13-CB
Single-mode 25Gbps
1310nm 10km


Migration to 100G
SFP25G
SFP25G
SFP25G
SFP25G

Multi-Port Copper Breakout

25GbE Consortium already has a 50GbE (2X25G) Interface Defined

- Breakout to single Lanes

  QSFPxx capable of running at 4X 10GbE, 25GbE or 50GbE
  – xx = +, 28 or 56

1 QSFP

4 SFP

To 10GbE, 25GbE or 50GbE SFP Ports

- Breakout to Dual Lanes

  QSFPxx capable of running at 2X50GbE, or 100GbE

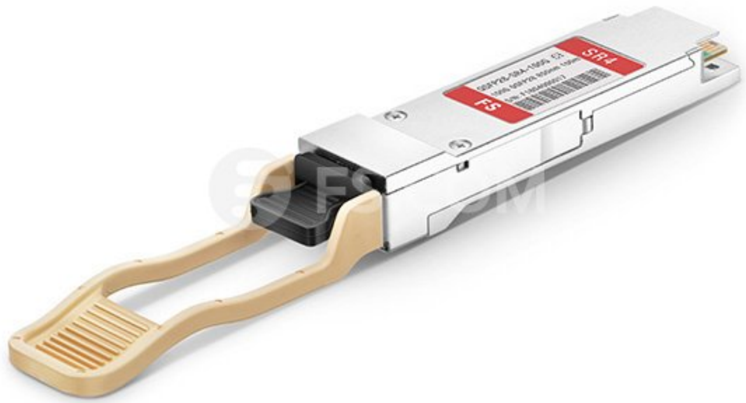
  – xx = +, 28 or 56

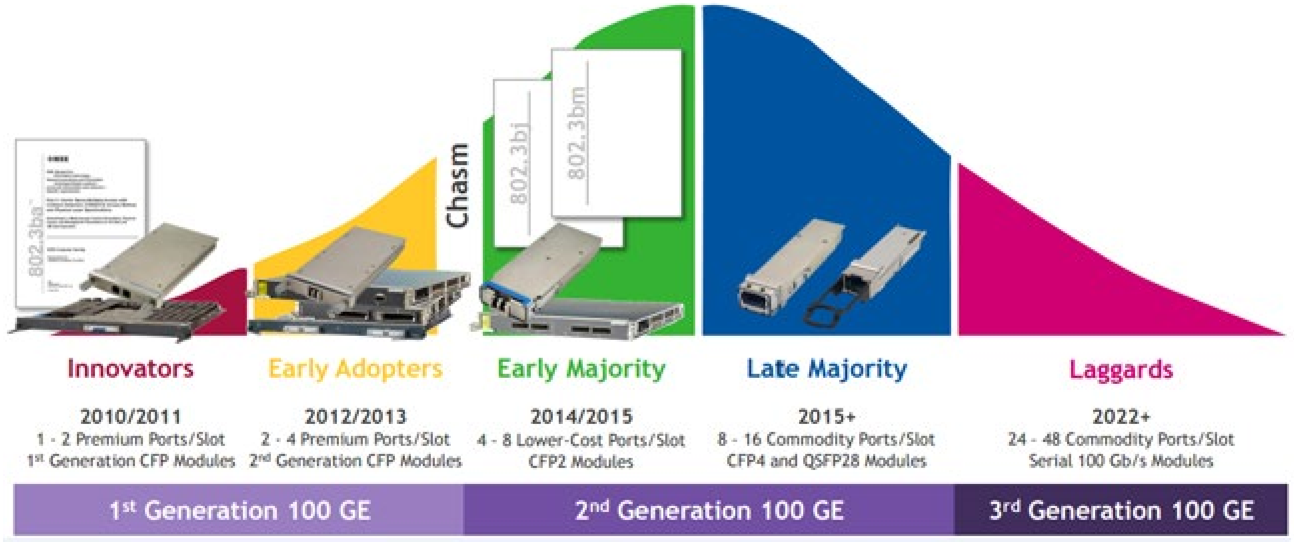
1 QSFP

2 QSFP

To 50GbE (2X25G) or 100GbE (2X50GbE) QSFP Ports

SERIAL DATA INTERFACES - CAUI-4, ROMAN NUMERAL FOR 100 IS C, 100G





To switch between 2.5G/10G and 1G, a RCW change is needed which requires a reboot of the SoC.

A “backdoor” SW sequence can avoid this HRESET if feasible. USXGMII eliminates this restriction.

NXP Semiconductors
Application Note

Document Number: AN5316
Rev. A, 08/2016

SGMII to OC-SGMII Speed Switching for the NXP T-Series Product Family

A Method to Override Serial Ethernet Protocol Selection Settings via Software

Contents

1 Introduction

Various Ethernet speeds are supported by the devices in the NXP QorIQ Communications Processors product family portfolio. One speed that is offered is 2.5 Gbps SGMII, or sometimes referred to as Over Clocked SGMII (OC-SGMII). However, this speed configuration is not designed to auto-negotiate to any other lower speed and is essentially fixed (2500 Base-X, for example). SGMII lane speed is chosen via a hardware Reset Configuration Word (RCW) selection.

Document Number	EDCS-1517762
Based on Template	EDCS-189229 Rev 20
Created By	Amrik Bains
Contributors	Arvind Kansal

Universal SXGMII PHY-MAC Interface for Single and Multiple Network Port

The Universal Serial Media Independent Interface for carrying multiple network ports over a single SERDES (USXGMII-M) is specified in this document to meet the following requirements:

- Convey Single and MULTIPLE network ports over an USXGMII MAC-PHY interface
- Utilize a 64/66 PCS to minimize power and serial bandwidth
- Use modified 802.3by section 108.5.2.4, to add Alignment Markers to support multiple ports over single SERDES
- System Interface operates in full duplex mode only
- Ability to send PTP time stamp from PHY to MAC to improve accuracy/jitter on encrypted/non-encrypted PTP packet with MACSec is in the ASIC
- Hardware assisted auto-negotiation for all supported speeds
- Flexibility to add new features using Extension Field in pre-amble

NBASE-T®

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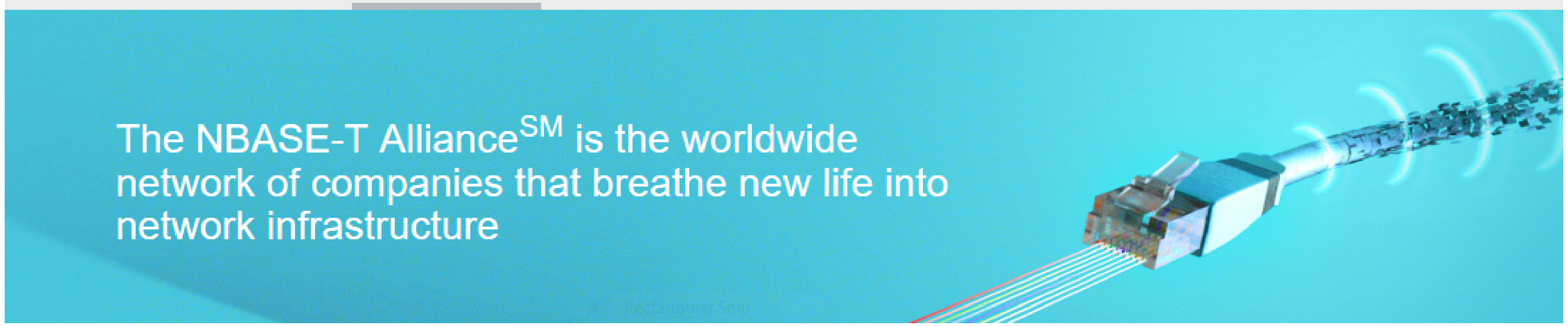


What is NBASE-T™ Technology

NBASE-T™ technology defines a new type of Ethernet signaling that boosts the speed of installed based twisted-pair cabling well beyond the cable's designed limit of 1 Gigabit per second (Gbps) for distances up to 100 meters.

Capable of reaching 2.5 and 5 Gbps using the large installed base of Cat5e and Cat6 cabling, NBASE-T solutions enable users to accelerate their networks in the most cost-effective, least disruptive manner.

TECHNOLOGY

- NBASE-T TECHNOLOGY
- WHAT IS NBASE-T™ TECHNOLOGY
- APPLICATIONS
- NBASE-T PRODUCTS

SoC	USXGMII Supported	SerDes Speed	SerDes RefClk	Number of Ports	Port Types
LX2160	10G-SXGMII	10.3125 Gbaud	156.25MHz	1	10G, 2.5G, 1G, 100M
LX2160	5G-SXGMII	5.15625 Gbaud	156.25MHz	1	2.5G, 1G, 100M
LS1028A	10G–SXGMII	10.3125 Gbaud	156.25MHz	1	2.5G, 1G, 100M, 10M
LS1028A	10G-QXGMII	10.3125 Gbaud	156.25MHz	4	2.5G, 1G, 100M, 10M

- USXGMII can operate at 100 Mbps, 1 Gbps, 2.5 Gbps, 5 Gbps or 10 Gbps (NXP not supporting 5Gbps on LX2)
  - for 10G-SXGMII, the serial link frequency is 10.3125GHz
  - for 5G-SXGMII, the serial link frequency is 5.15625 GHz
- e.g., The lower effective data rates are implemented by replicating the data for 10G-SXGMII
  - 2 times for 5 Gbps
  - 4 times for 2.5 Gbps
  - 5 times for 2 Gbps,
  - 10 times for 1 Gbps,
  - 50 times for 200 Mbps
  - 100 times for 100 Mbps
  - 1000 times for 10 Mbps

- A standard that defines 2.5 and 5 Gigabit Ethernet speeds over the large installed base of Cat5e and Cat6 cables (2.5GBASE-T and 5GBASE-T) in the Enterprise and SMB infrastructure.
- The standard offers 5x faster speed than legacy Gigabit Ethernet with no changes to the cabling infrastructure.

IEEE Std 802.3bz™-2016

(Amendment to IEEE Std 802.3™-2015
as amended by IEEE Std 802.3bw™-2015, IEEE Std 802.3by™-2016, IEEE Std 802.3bq™-2016, IEEE Std 802.3bp™-2016,
IEEE Std 802.3br™-2016, and IEEE Std 802.3bn™-2016)

IEEE Standard for Ethernet

Amendment 7: Media Access Control Parameters, Physical Layers, and Management Parameters for 2.5 Gb/s and 5 Gb/s Operation, Types 2.5GBASE-T and 5GBASE-T

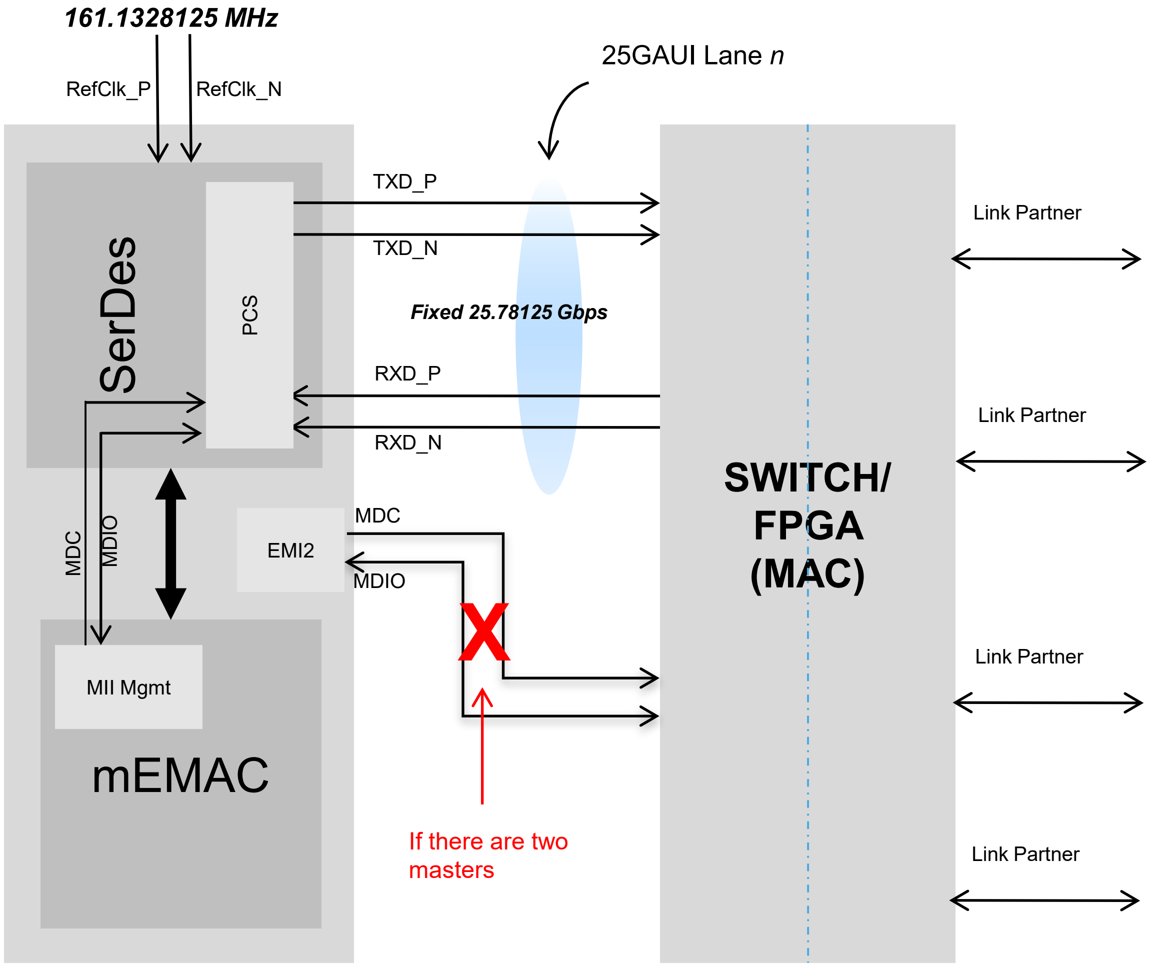
LAN/MAN Standards Committee
of the
IEEE Computer Society

Approved 22 September 2016
IEEE-SA Standards Board

PHYless Considerations

SECURE CONNECTIONS
FOR A SMARTER WORLD

- SerDes block is used (differential signaling)
- 25Gbps supported with SerDes lane operating at 25.78125 Gbps
- MDIO for external and internal buses
- May lose MII management because no slave device. Configuration and status info can be placed in system memory for access or maybe SPI or I2C (depends on the connected device).



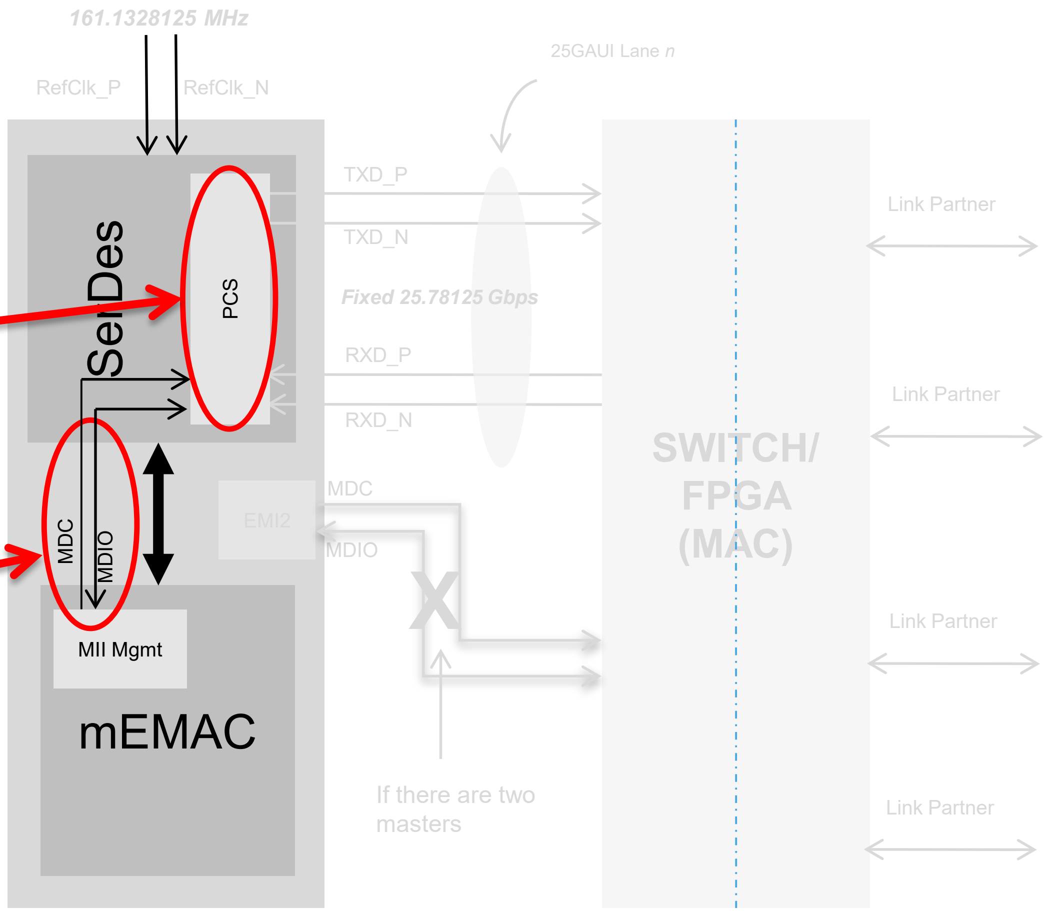
MII Management Bus and Controller

SECURE CONNECTIONS
FOR A SMARTER WORLD

Ethernet Management Interface 1					
EMI1_MDC	Management Data Clock	AF2	O	OV_{DD}	---
EMI1_MDIO	Management Data In/Out	AE1	IO	OV_{DD}	---
Ethernet Management Interface 2					
EMI2_MDC	Management Data Clock	AG1	O	OV_{DD}	---
EMI2_MDIO	Management Data In/Out	AG3	IO	OV_{DD}	5, 6

- OV_{DD} = 1.8V
- [5,6]EMI2_MDIO is open-drain (requires weak pull-up)
- Dedicated controllers for external devices
- Each MAC has internal bus to on-chip PCS per port

- MDC is typically 2.5MHz. Programmable in MDIO Register MDIO_CFG.
- 25G PCS has Status and Control registers
- Internal MDIO bus access per MAC. MDEV_PORT defines the Port Address.
- Clause 45 protocol for 25G PCS



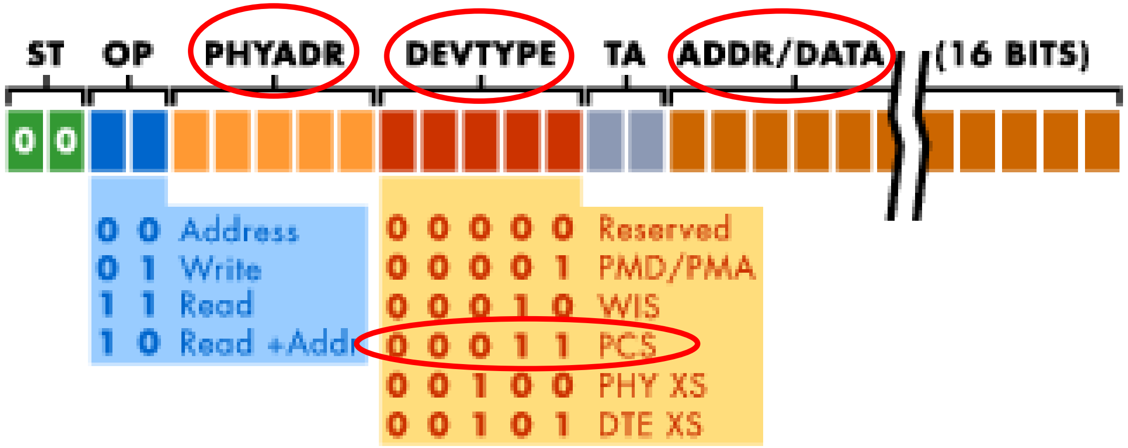
TIP: Each MAC has only 1 internal device on the bus. Ensures no addressing conflicts.

- Clause 45 requires three arguments:
  - PORT
  - DEVICE
  - REGISTER ADDRESS

- E25GaCR1/ANLTmCR1[MDEV_PORT] defines the PCS PORT address for Clause 45 accesses. (LX2 *SoC Reference Manual*)

- MDIO_CTL, MDIO_DATA, & MDIO_ADDR used to perform reads/writes to specific registers. (LX2 *DPAA Reference Manual*)

In order to address the deficiencies of Clause 22, Clause 45 was added to the 802.3 specification. Clause 45 added support for low voltage devices down to 1.2V and extended the frame format to provide access to many more devices and registers. Some of the elements of the extended frame are similar to the basic data frame:



ST	2 bits	Start of Frame (00 for Clause 45)
OP	2 bits	OP Code
PHYADR	5 bits	PHY Address
DEVTYPE	5 bits	Device Type
TA	2 bits	Turnaround time to change bus ownership from STA to MMD if required
ADDR/DATA	16 bits	Address or Data Driven by STA for address Driven by STA during write Driven by MMD during read Driven by MMD during read-increment-address

Offset 0x030

Access: Mixed

Bit	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
R	BSY	CMP	CIM	—					NEG[1]	EHOLD	—					
W		w1c														
Resets: Internal	All zeros															
External	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0

Bit	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
R	MDIO_CLK_DIV[2]									ENC45	PRE	MDIO_HOLD			MDIO_RD_ER	—
W																
Resets: Internal	0	0	0	1	0	1	0	0	0	1	0	0	1	0	0	0
External	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0

Figure 6-93. MDIO_CFG Register Definition

1 Reset value of NEG for external MDIOs is 1

2 Reset value of MDIO_CLK_DIV for external MDIOs is 9'b0

TIP: Ensure MDIO_CFG[ENC45]=0b1 when accessing 25G PCS resisters.

MDIO CONTROL REGISTER (MDIO_CTL) ALSO USED FOR CLAUSE 22 FIVE BIT REGISTER ADDRESSES.

Offset 0x034

Access: Read/Write

Bit	0–15
R/W	—
Reset	All zeros

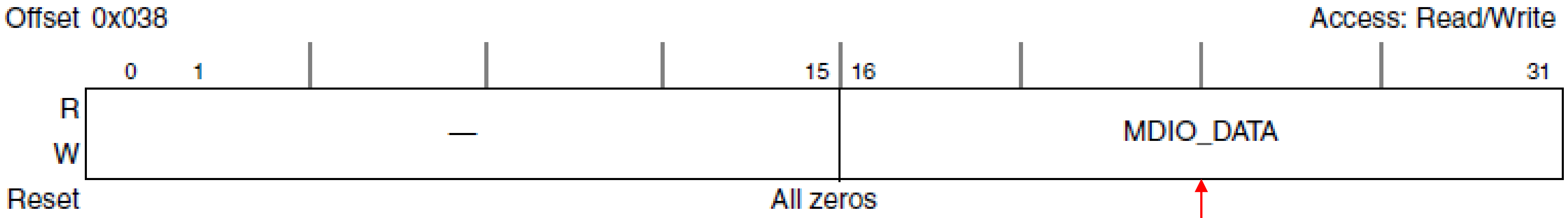
Bit	16	17	18–21	22–26	27–31
R/W	READ	POST_INC	—	PORT_ADDR/PHY_ADDR	DEV_ADDR/REG_ADDR
Reset			0x0001		

TIP: To ensure a data read does not contain stale information, a 0 -> 1 transition on bit 16 is recommended

TIP:
For Clause 45, this is the PORT address.
For Clause 22, this is the PHY address.

TIP:
For Clause 45, this is the DEVICE address
For Clause 22, this is the REGISTER address

MDIO DATA REGISTER (MDIO_DATA) – 16 BIT DATA REGISTER



TIP:
For reads, contains results
For writes, contains info to program (sent when written)

MDIO CLAUSE 45 ADDRESS REGISTER (REGADDR) – 32 BITS

Offset 0x03C | Access: Write only

	0 … 31
R	
W	REGADDR
Reset	All zeros

TIP: FOR CLAUSE 45 ONLY

26.5.1.1 PCVTR 25G_100G Devices Address

This table describes device address for different divces.

Table 26-4. Device Address Overview

Device	Address
PCS	3
RS-FEC	30
AN-LT	7

Chapter 26 SerDes Module

Offset REGADDR	Register e.g., DEVICE ADDR = 0x3, PCS MDIO Registers	Width (In bits)	Access	Reset value
0h	CONTROL_1 (CONTROL_1)	16	RW	2040h
1h	PCS status 1 (STATUS_1)	16	RO	0002h
2h	DEVICE_ID0 (DEVICE_ID0)	16	RO	0000h
3h	DEVICE_ID1 (DEVICE_ID1)	16	RO	0000h

26.5.2.1.1.2 PCS status 1 (STATUS_1)

26.5.2.1.1.2.1 Offset

DEVICE ADDR = 3

Register	Offset REGADDR = 1
STATUS_1	1h

26.5.2.1.1.2.3 Diagram

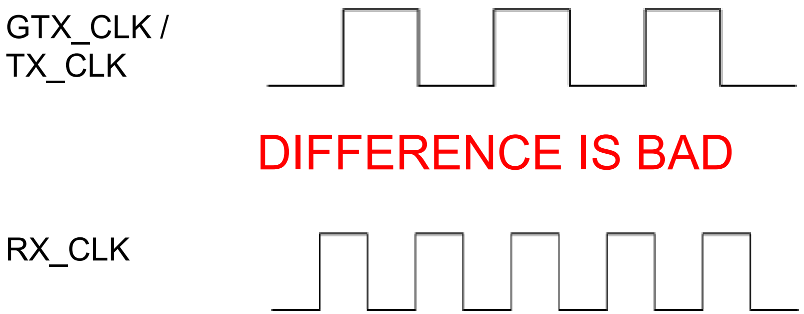
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved								Fault	Reserved				PCS_Receive_Link_Status	Low_power_Ability	Reserved
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0

TIP: Register 0x3.0x20 also has link status information.
Per 802.3, 3.1.2 is just a latch-low version of 3.32.12

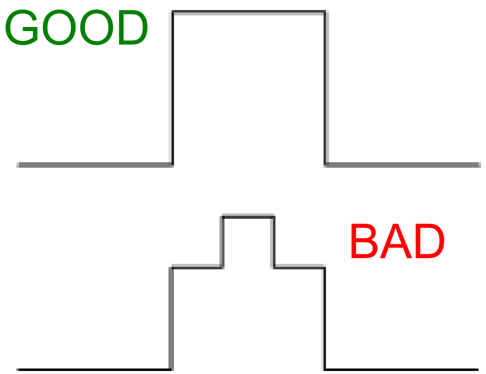
Signal Integrity Considerations

SECURE CONNECTIONS
FOR A SMARTER WORLD

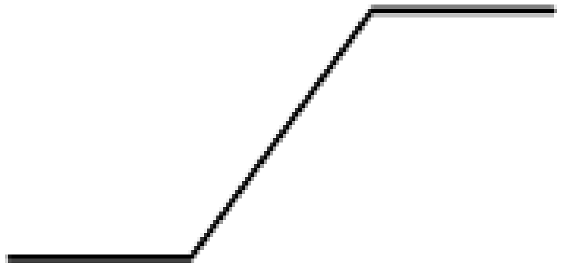
- FREQ Greater than +/- 300 PPM



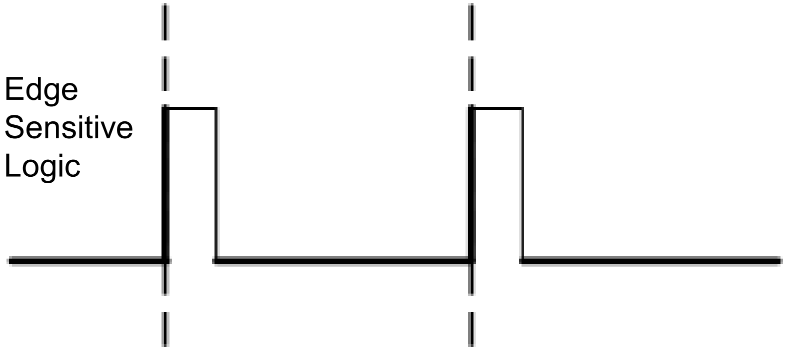
- Monotonic Edges

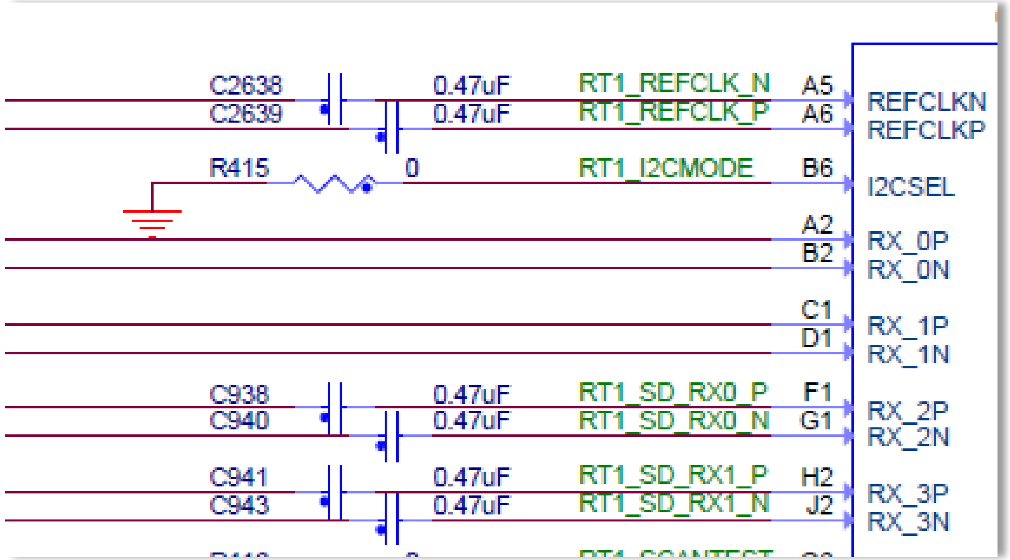


- Rise/Fall Meets Spec



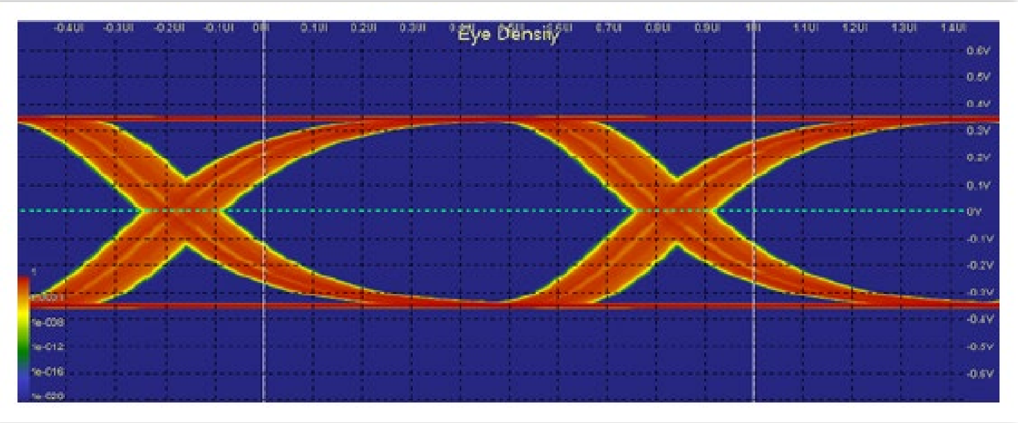
- Really Duty Cycle Agnostic



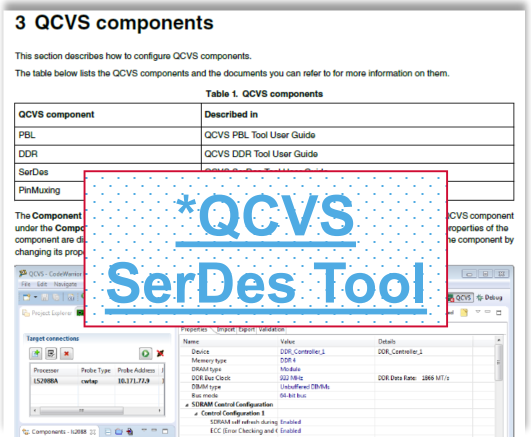


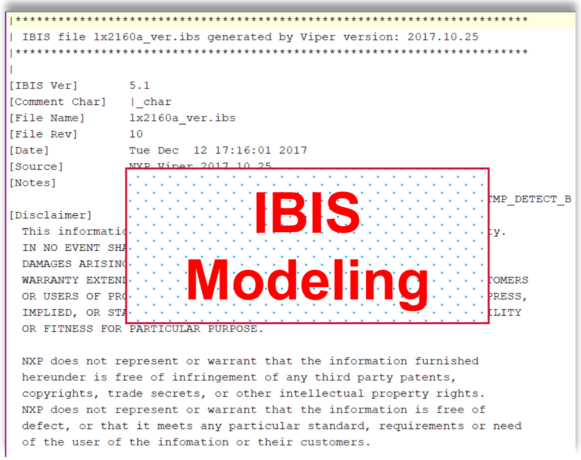
A/C Coupling Capacitors

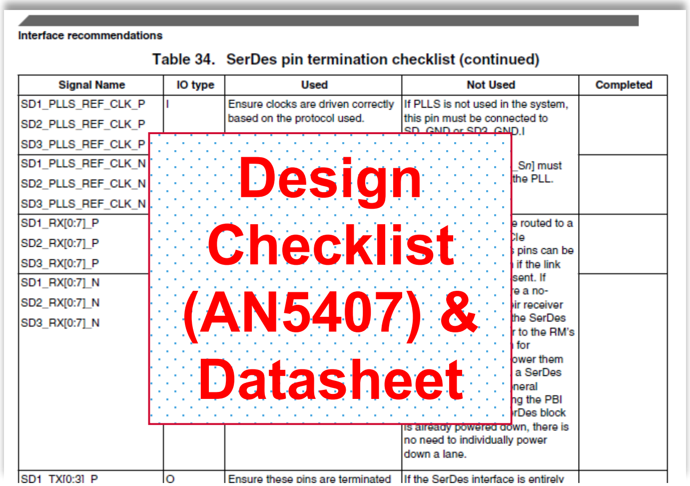
The Management Complex (MC) firmware automatically applies the tuned transmit equalization settings. Check version of MC firmware.

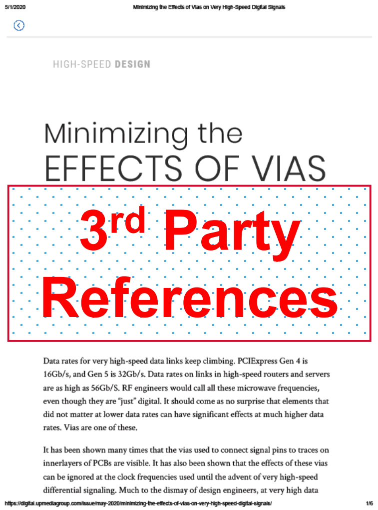


SIGNAL INTEGRITY CONSIDERATIONS – OTHER RESOURCES









Documentation : SoC and DPAA2 Reference Manuals

SECURE CONNECTIONS FOR A SMARTER WORLD

SoC
RM for SerDes & PCS info

26.4.1.66.2 Function

E25GaCR1 contains control bits used for the E25Ga protocol

26.4.1.66.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R/W	MDEV_PORT					Reserved										
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R/W	Reserved															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

DPAA
RM for MAC info

12.4.1 mEMAC Top-Level Memory Map

The Ethernet MAC controller and the Ethernet management interface is allocated 2 Kbytes of memory-mapped space, as indicated in Table 12-3 .

Table 12-3. Ethernet MAC Memory Map Summary

Address Offset	Function
0x000–0x02C	General control and status registers
0x030–0x03C	MII management registers
0x040–0x0E8	General control and status registers
0x100–0x1FC	Rx Statistics counters registers
0x200–0x2FC	Tx Statistics counters registers
0x380-0x3FC	PFC Statistics counters registers

Development Boards

SECURE CONNECTIONS
FOR A SMARTER WORLD

Board features

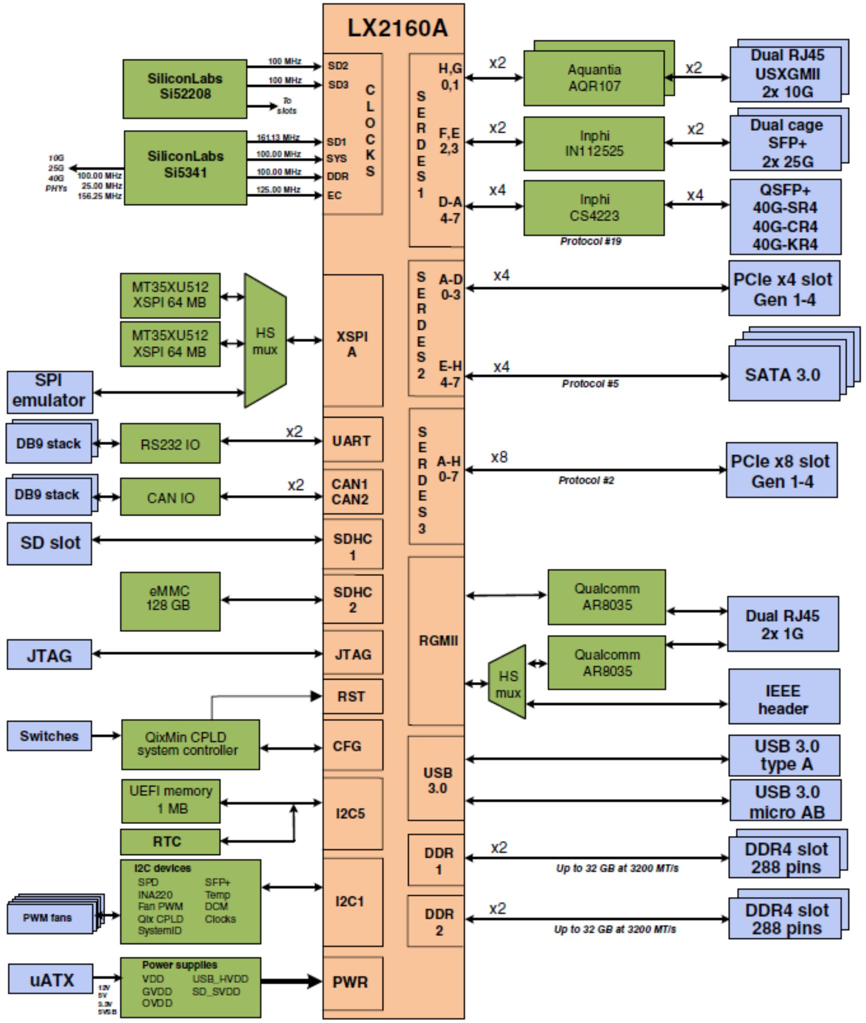


Figure 1-2. LX2160ARDB block diagram

Document Number: LX2160ARDBRM

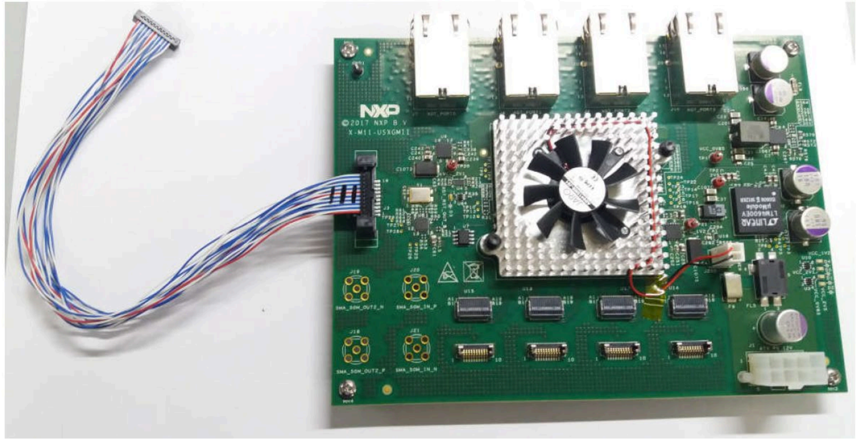
Rev. 1, 10/2018

Table 2-12. Devices supported on mezzanine cards

Mezzanine card	Purpose	Major peripheral devices
X-M1-PCIE-X8 (M1)	Test PCIe Gen 1/2/3/4 protocol on x1/x2/x4/x8 lanes; test SGMII protocol	One PCIe x16 slot with an EMI header (for SGMII support)
X-M4-PCIE-SGMII (M4)	Test PCIe Gen 1/2/3/4 protocol on x1/x2/x4 lanes; test SGMII protocol	One PCIe x8 slot with an EMI header (for SGMII support)
X-M5-SATA (M5)	Test SATA protocol	Four SATA headers
X-M6-PCIE-X8 (M6)	Test PCIe Gen 1/2/3 protocol on x1/x2/x4/x8 lanes	One PCIe x8 slot with a PCIe hot plug controller
X-M7-40G (M7)	Test 40 Gbit Ethernet	1x4 SFP+ cage assembly, with each cage supporting a 10 GbE transceiver
X-M8-100G (M8)	Test 100 Gbit Ethernet	One QSFP28 cage to support a 100 GbE transceiver
X-M11-USXGMII (M11)	Test USXGMII protocol	Four RJ45 connectors
X-M12-XFI (M12)	Test XFI protocol	1x4 SFP+ cage assembly, with each cage supporting an XFI-compliant 10 GbE transceiver
X-M13-25G (M13)	Test 25/50 Gbit Ethernet	1x4 SFP28 cage assembly, with each cage supporting a 25 GbE transceiver

Document Number: LX2160AQDSRM
Rev. D, 09/2018

X-M11-USXGMII(29828)



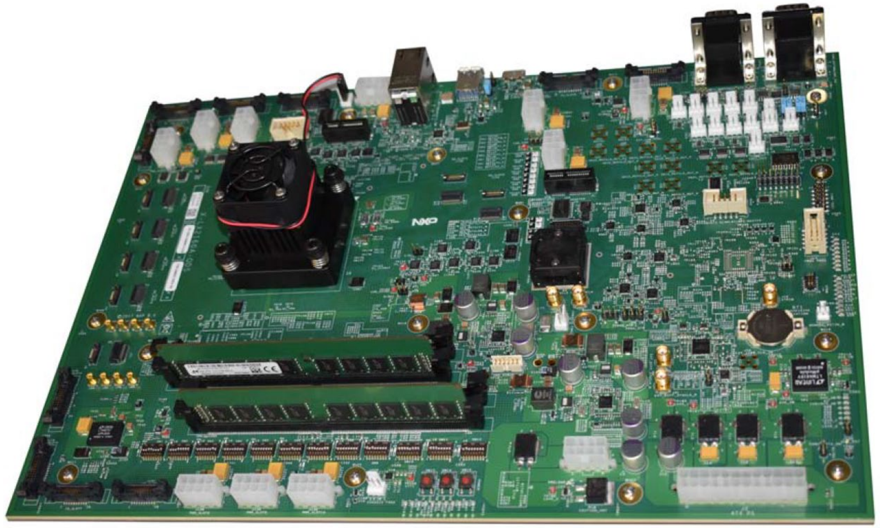


Table 12. Mezzanine cards (continued)

Mezzanine card	Purpose	Supported SerDes connectors on QDS board	Cables for SerDes connection
X-M7-40G (M7)	Test 40 Gbit Ethernet	J108 and J111	Samtec type 1 cable assembly (HDR-198816-XX-ECUE)
X-M11-USXGMII (M11)	Test USXGMII protocol	J108, J111, and J117	Samtec type 2 cable assembly (HDR-198564-01-ECUE)
X-M12-XFI (M12)	Test XFI protocol	J108, J111, and J117	Samtec type 2 cable assembly (HDR-198564-01-ECUE)

You need to put the PCS into KX mode. Clause 45 is only available in KX mode and Clause 22 is only available in SGMII mode. The default is SGMII.

As documented in the RM section 31.8.1.3:
After initializing the MAC and SERDES lane(s), and prior to initiating any 1000Base-KX traffic, perform the following 1000Base-KX protocol initialization:
1. Enable the 1000Base-KX AN reference clock by setting PLLnCR0[DLYDIV_SEL]=01, for the PLLn that is the clock source for the SGMII PCS (see TPLL_LES in General Control Register 0 - Lane a (LNAGCR0 - LNDGCR0)).
2. Enable the 1000Base-KX AN module by setting PCCR8[SGMIIp_KX]=1, for each SGMIIp that will run in 1000Base-KX rather than SGMII mode.
3. Initialize SGMII IF Mode register to 0x0008:
• SGMII_EN=0
• USE_SGMII_AN=0
• SGMII_SPEED=10
5. Initialize 1000Base-KX (Clause 45) AN Advertisement Register 1:
• TX_NONCE=unique value per device
6. Read 1000Base-KX (Clause 45) AN Status Register to clear any previous status
7. Initialize 1000Base-KX (Clause 45) AN Control Register to 0x1200:
• AN_ENAB=1
• RST_AN=1

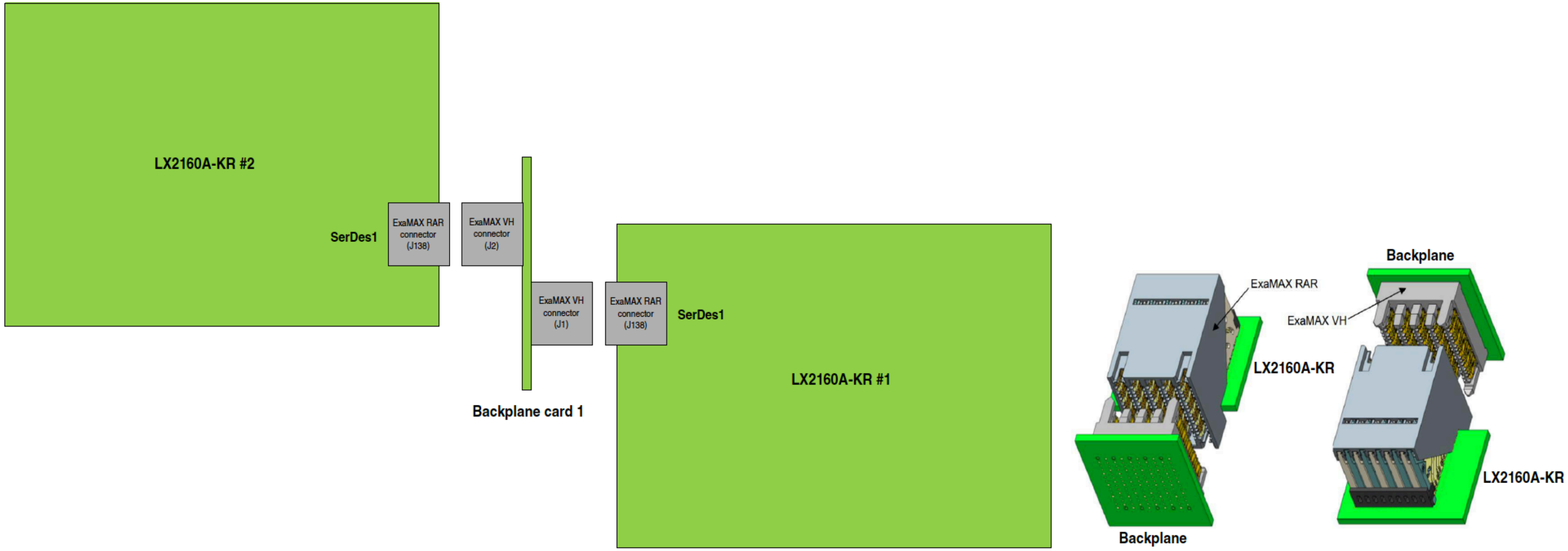


Figure 3. KR-to-KR connection with backplane card 1

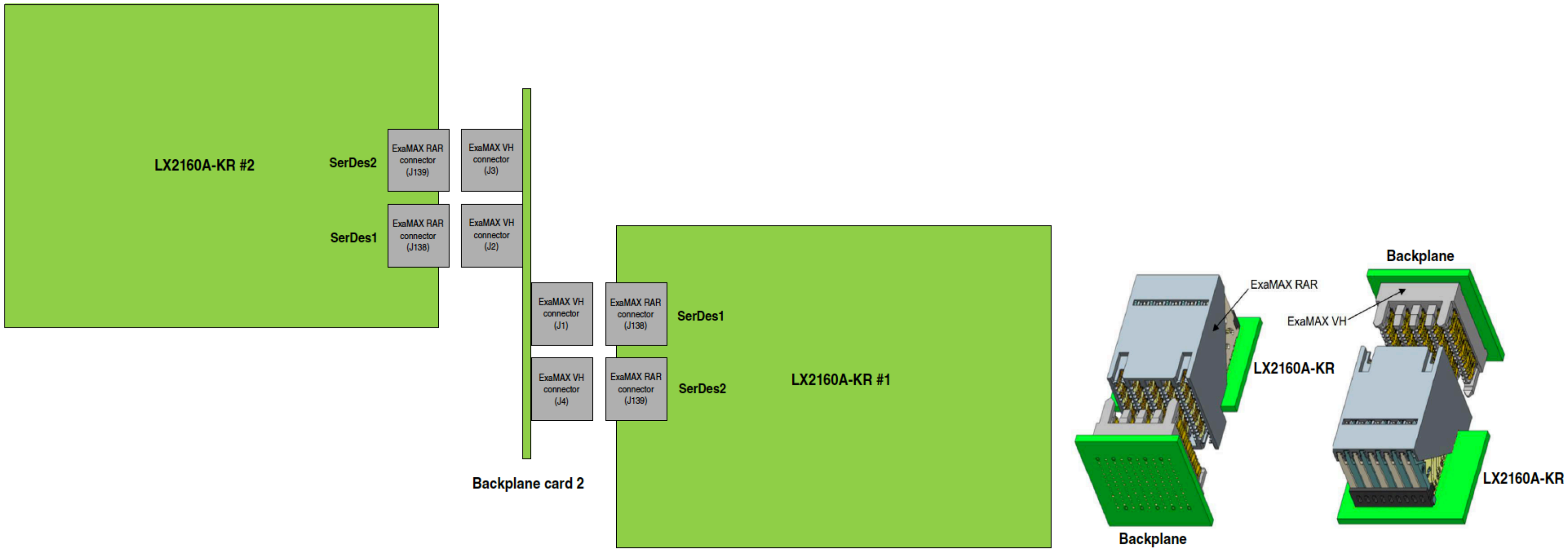


Figure 4. KR-to-KR connection with backplane card 2

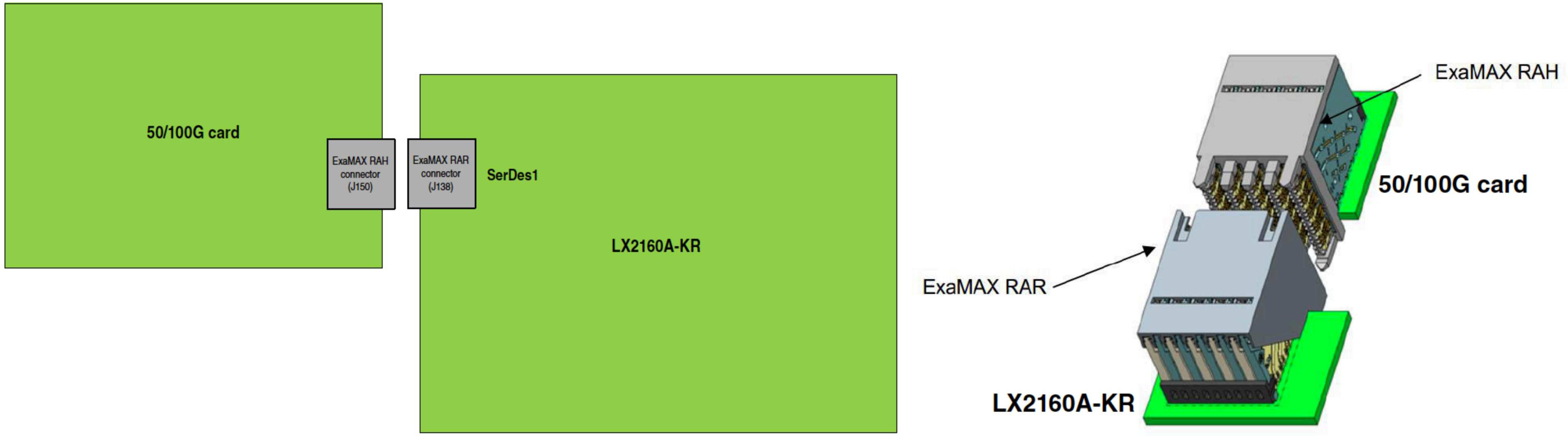


Figure 5. KR board to 50/100G card connection

U-Boot command line check for link status

SECURE CONNECTIONS
FOR A SMARTER WORLD

#Read 0x3.0x1

clearing the READ bit and setting DEVICE=0x3 in MDIO_CTL
=> mw.l 08c0f**034** 03
***
choosing register 0x1 with REGADDR
=> mw.l 08c0f**03c** 01
***
performing first read of 3.1
=> mw.l 08c0f**034** 8003
clearing the READ bit and setting DEVICE=0x3 in MDIO_CTL
=> mw.l 08c0f**034** 03
***
performing second read of 3.1
=> mw.l 08c0f**034** 8003
***
display returned data in MDIO_DATA
=> md 08c0f**038** 1
08c0f038: 0000000**4** Compare with bit #2 in register defined in slide #34

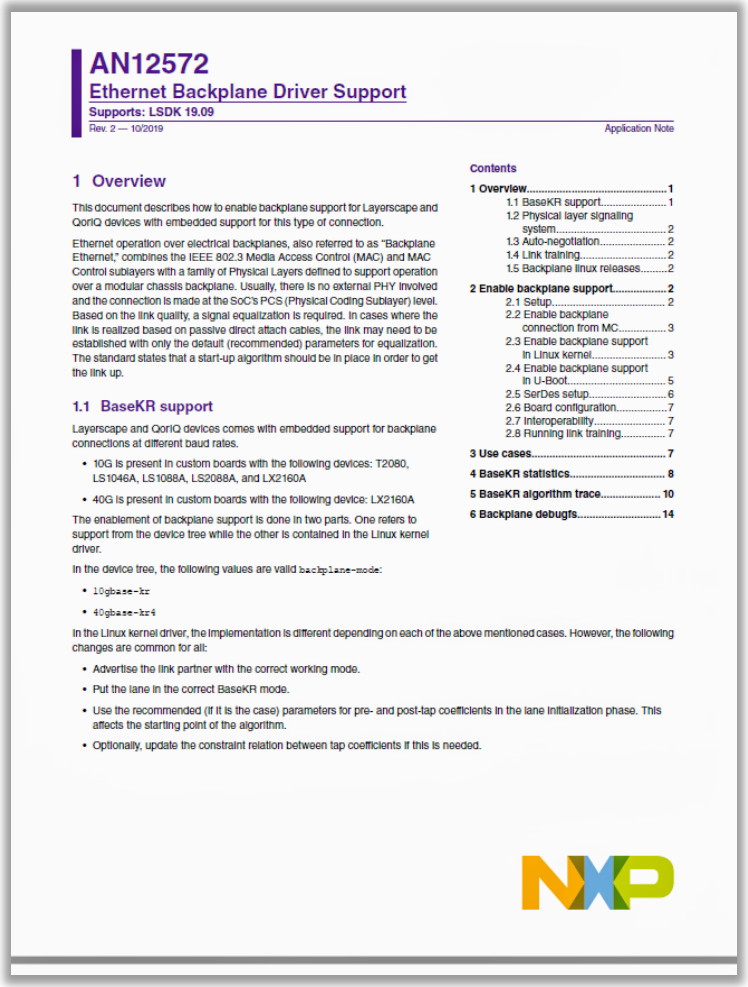
Summary / Conclusion

SECURE CONNECTIONS
FOR A SMARTER WORLD

- *Data Path (PCS, Interfaces and MACs)*
  - Internal PCS Link Status should follow the live link (reception of valid symbols should set link status bit)
    - e.g., unplug cable or power down lane ***should*** clear link bit (assuming logical idles are not still being received by the LX2)
  - Consult Design Checklist (AN5407) for unused pin termination recommendations. Floating inputs can cause undesired behavior. A/C coupling capacitors are recommended.
  - Minimum Platform Frequency for 25GbE: 645MHz
  - CDR Lock, PCS Link-up, and no faults are good indicators data can TX & RX
  - Spread-spectrum clocking is for the PCI Express protocol only
  - PCB trace length is not the only factor in High Speed Serial Interface (HSSI) design. IBIS simulation is key.
  - Ensure electrical specifications are met
  - Check errata doc for any known issues

- *Management Path (Controllers for External and Internal Buses)*
  - MDIO_CFG controls MDC frequency
  - 3 parameters to access internal PCS Clause 45 registers: port, device, & register
    - MDC/MDIO bus transactions using mEMAC memory-mapped registers
  - Consult Design Checklist (AN5407) for unused pin termination recommendations. Floating inputs can cause undesired behavior. Also, confirm if MDC/MDIO pins are open-drain and may require a pull-up resistor
  - Ensure Electrical specifications are met
  - Check errata doc for any known issues

SUMMARY / CONCLUSION – 2 OF 2



AN12572
Ethernet Backplane Driver Support
Supports: LSDK 19.09

Rev. 2 — 10/2019 Application Note

Contents

1 Overview

This document describes how to enable backplane support for Layerscape and QorIQ devices with embedded support for this type of connection.

Ethernet operation over electrical backplanes, also referred to as "Backplane Ethernet," combines the IEEE 802.3 Media Access Control (MAC) and MAC Control sublayers with a family of Physical Layers defined to support operation over a modular chassis backplane. Usually, there is no external PHY involved and the connection is made at the SoC's PCS (Physical Coding Sublayer) level. Based on the link quality, a signal equalization is required. In cases where the link is realized based on passive direct attach cables, the link may need to be established with only the default (recommended) parameters for equalization. The standard states that a start-up algorithm should be in place in order to get the link up.

1.1 BaseKR support

Layerscape and QorIQ devices comes with embedded support for backplane connections at different baud rates.

- 10G is present in custom boards with the following devices: T2080, LS1046A, LS1088A, LS2088A, and LX2160A
- 40G is present in custom boards with the following device: LX2160A

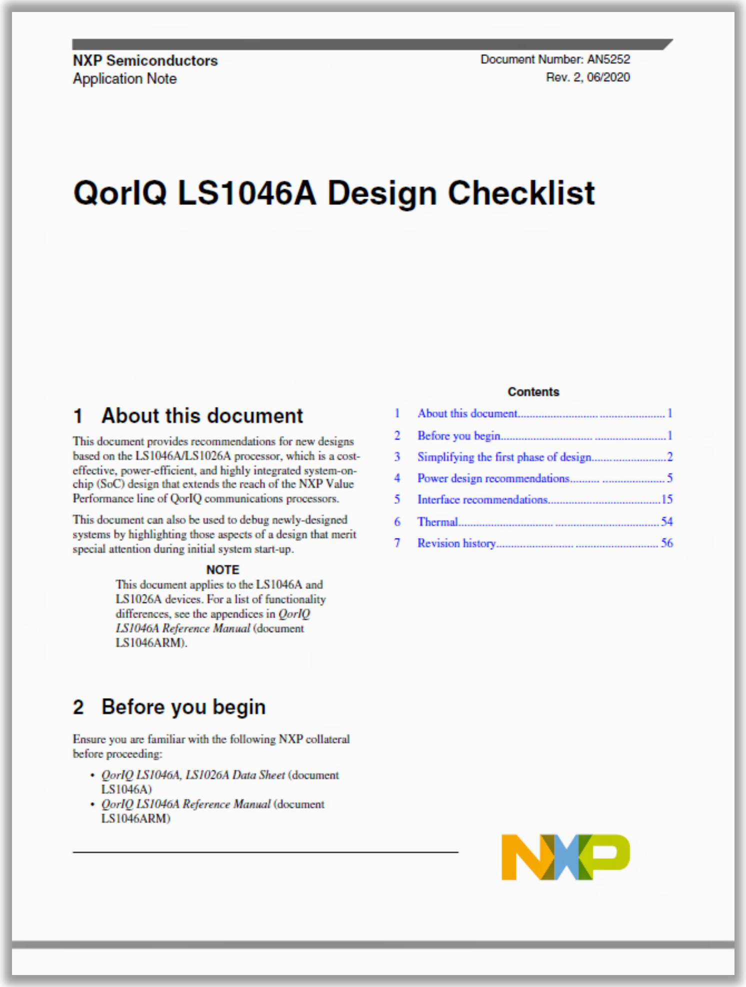
The enablement of backplane support is done in two parts. One refers to support from the device tree while the other is contained in the Linux kernel driver.

In the device tree, the following values are valid `backplane-mode`:

- `10gbase-kr`
- `40gbase-kr4`

In the Linux kernel driver, the implementation is different depending on each of the above mentioned cases. However, the following changes are common for all:

- Advertise the link partner with the correct working mode.
- Put the lane in the correct BaseKR mode.
- Use the recommended (if it is the case) parameters for pre- and post-tap coefficients in the lane initialization phase. This affects the starting point of the algorithm.
- Optionally, update the constraint relation between tap coefficients if this is needed.



NXP Semiconductors Document Number: AN5252
Application Note Rev. 2, 06/2020

QorIQ LS1046A Design Checklist

1 About this document

This document provides recommendations for new designs based on the LS1046A/LS1026A processor, which is a cost-effective, power-efficient, and highly integrated system-on-chip (SoC) design that extends the reach of the NXP Value Performance line of QorIQ communications processors.

This document can also be used to debug newly-designed systems by highlighting those aspects of a design that merit special attention during initial system start-up.

NOTE

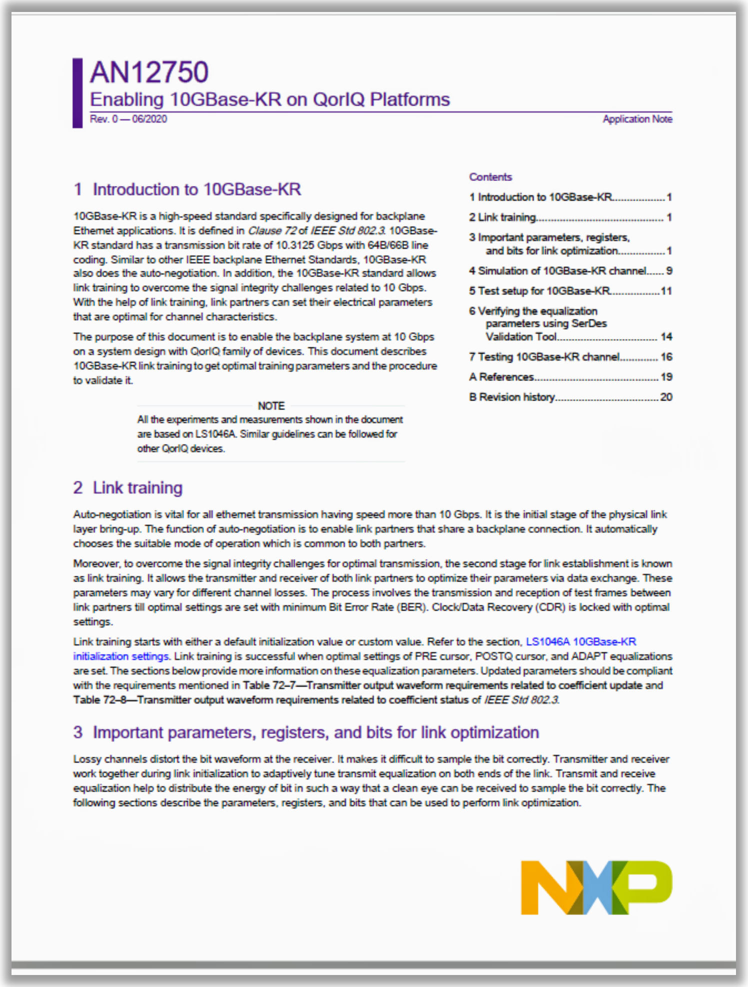
This document applies to the LS1046A and LS1026A devices. For a list of functionality differences, see the appendices in *QorIQ LS1046A Reference Manual* (document LS1046ARM).

Contents

2 Before you begin

Ensure you are familiar with the following NXP collateral before proceeding:

- *QorIQ LS1046A, LS1026A Data Sheet* (document LS1046A)
- *QorIQ LS1046A Reference Manual* (document LS1046ARM)



AN12750
Enabling 10GBase-KR on QorIQ Platforms
Rev. 0 — 06/2020 Application Note

Contents

1 Introduction to 10GBase-KR

10GBase-KR is a high-speed standard specifically designed for backplane Ethernet applications. It is defined in *Clause 72* of *IEEE Std 802.3*. 10GBase-KR standard has a transmission bit rate of 10.3125 Gbps with 64B/66B line coding. Similar to other IEEE backplane Ethernet Standards, 10GBase-KR also does the auto-negotiation. In addition, the 10GBase-KR standard allows link training to overcome the signal integrity challenges related to 10 Gbps. With the help of link training, link partners can set their electrical parameters that are optimal for channel characteristics.

The purpose of this document is to enable the backplane system at 10 Gbps on a system design with QorIQ family of devices. This document describes 10GBase-KR link training to get optimal training parameters and the procedure to validate it.

NOTE

All the experiments and measurements shown in the document are based on LS1046A. Similar guidelines can be followed for other QorIQ devices.

2 Link training

Auto-negotiation is vital for all ethernet transmission having speed more than 10 Gbps. It is the initial stage of the physical link layer bring-up. The function of auto-negotiation is to enable link partners that share a backplane connection. It automatically chooses the suitable mode of operation which is common to both partners.

Moreover, to overcome the signal integrity challenges for optimal transmission, the second stage for link establishment is known as link training. It allows the transmitter and receiver of both link partners to optimize their parameters via data exchange. These parameters may vary for different channel losses. The process involves the transmission and reception of test frames between link partners till optimal settings are set with minimum Bit Error Rate (BER). Clock/Data Recovery (CDR) is locked with optimal settings.

Link training starts with either a default initialization value or custom value. Refer to the section, LS1046A 10GBase-KR initialization settings. Link training is successful when optimal settings of PRE cursor, POSTQ cursor, and ADAPT equalizations are set. The sections below provide more information on these equalization parameters. Updated parameters should be compliant with the requirements mentioned in Table 72–7—Transmitter output waveform requirements related to coefficient update and Table 72–8—Transmitter output waveform requirements related to coefficient status of *IEEE Std 802.3*.

3 Important parameters, registers, and bits for link optimization

Lossy channels distort the bit waveform at the receiver. It makes it difficult to sample the bit correctly. Transmitter and receiver work together during link initialization to adaptively tune transmit equalization on both ends of the link. Transmit and receive equalization help to distribute the energy of bit in such a way that a clean eye can be received to sample the bit correctly. The following sections describe the parameters, registers, and bits that can be used to perform link optimization.

SECURE CONNECTIONS
FOR A SMARTER WORLD

Terminology

- SoC: System On a Chip
- DPAA2: Data Path Acceleration Architecture, second version
- MC: Management Complex
- LLC: Link Layer Control
- MAC: Media Access Controller
- MII: Media Independent Interface
- RS: Reconciliation Layer
- PCS: Physical Coding Sublayer
- SFP: Small Form/Factor Pluggable
- PHY: Physical Transceiver
- WRIOP: Wire rate IO Processor
- FMAN: Frame Manager
- eTSEC: Enhanced Three-Speed Ethernet Controller
- PPFE: Packet Processing Forwarding Engine
- mEMAC: Multi-rate Ethernet MAC
- cEMAC: 100G Multi-rate Ethernet MAC
- RCW: Reset Configuration Word
- RS-FEC: Reed-Solomon Forward Error Correction
- FC-FEC: Firecode Forward Error Correction
- IBIS: Input/Output Buffer Information Specification
- QCVS:QorIQ Configuration and Validation Suite
- AUI: Attachment Unit Interface
- PHY: Physical Transceiver